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A Numerical Study of Ground-State Boson Stars with Quartic Self-Interaction

A Bachelor's Thesis by

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Περίληψη

Η παρούσα πτυχιακή εργασία παρουσιάζει μια ολοκληρωμένη αριθμητική μελέτη αστέρων μποζονίων θεμελιώδους κατάστασης. Οι αστέρες μποζονίων είναι διαμορφώσεις βαθμωτών πεδίων ϕ σε ισορροπία και αυτο-αλληλεπιδρούν με απωστικό δυναμικό τέταρτης τάξης της μορφής $\frac{1}{4}\lambda|\phi|^4$. Αυτά τα θεωρητικά, εξωτικά συμπαγή αντικείμενα προσφέρουν μια πιθανή εξήγηση για τη σκοτεινή ύλη ή θα μπορούσαν να μιμηθούν τη συμπεριφορά μελανών οπών. Αρχικά, χρησιμοποιούμε λογισμό μεταβολών στο πλαίσιο της Γενικής Σχετικότητας για να εξάγουμε τις εξισώσεις Einstein-Klein-Gordon που περιγράφουν το σύστημα. Στη συνέχεια, λύνουμε αριθμητικά το συνοριακό πρόβλημα ιδιοτιμών που προκύπτει με δύο μεθόδους: τον ενσωματωμένο επιλυτή `solve_bvp` της Python και τη μέθοδο σκόπευσης (shooting method).

Μια σύγκριση μεταξύ των δύο αριθμητικών προσεγγίσεων δείχνει ότι, παρόλο που τα αποτελέσματά τους βρίσκονται σε εξαιρετική συμφωνία, κάθε μέθοδος προσφέρει διαφορετικά πλεονεκτήματα. Ενώ η μέθοδος σκόπευσης (shooting method) εξασφαλίζει σύγκλιση με ελάχιστη ρύθμιση για μεμονωμένα προφίλ, η επίλυση με την `solve_bvp` έχει πολύ καλύτερη απόδοση, εκτελώντας εκτενείς σαρώσεις παραμέτρων πάνω από 50 φορές ταχύτερα. Τα φυσικά προφίλ που προκύπτουν από τα αποτελέσματα αυτών των μεθόδων αποκαλύπτουν ότι η αύξηση του αδιάστατου κεντρικού βαθμωτού πεδίου, σ_c , παράγει πιο συμπαγείς αστέρες. Αντίθετα, αυξάνοντας την ισχύ της αυτο-αλληλεπίδρασης, Λ , επεκτείνεται η ενεργή περιοχή του βαθμωτού πεδίου και μπορεί να υποστηρίξει μεγαλύτερες συνολικές μάζες. Αποτυπώνοντας τη σχέση μεταξύ της αδιάστατης μάζας, $\mathcal{M}$, και του κεντρικού πεδίου, σ_c , για κάθε οικογένεια διαμορφώσεων μπορούμε να προσδιορίσουμε τα χαρακτηριστικά όρια σταθερότητας. Για μη-αλληλεπιδρώντες "mini" αστέρες μποζονίων, υπολογίζουμε τη μέγιστη συνολική τους μάζα ως $M_{\max} = 0.633 [M_{\text{Pl}}^2/m]$ για τιμή κεντρικού πεδίου $\sigma_c = 0.271$.

Επιπρόσθετα, συγκεντρώνοντας ένα σύνολο δεδομένων για την αδιάστατη μέγιστη συνολική μάζα $\mathcal{M}_{\max}$, και την αντίστοιχη ισχύ της αυτο-αλληλεπίδρασης, Λ , εφαρμόζουμε γραμμική παλινδρόμηση για να εξάγουμε την εμπειρική σχέση $\mathcal{M}_{\max} = \sqrt{0.047\Lambda + 0.375}$. Αυτή η σχέση ανάγεται με ακρίβεια στην αναλυτική έκφραση $\mathcal{M}_{\max} \approx 0.22\Lambda^{1/2}$ για μεγάλες τιμές του Λ ($\Lambda \gg 1$).

Τέλος, ταξινομούμε τα τρία εξελικτικά τελικά στάδια των διαμορφώσεων με βάση τη δυναμική και ενεργειακή τους σταθερότητα: σταθερές δεσμευμένες διαμορφώσεις, ασταθείς δεσμευμένες διαμορφώσεις που καταρρέουν σε μελανές οπές, και ασταθείς αδέσμευτες διαμορφώσεις που διασκορπίζονται σε ελεύθερα σωματίδια στο άπειρο.

Abstract

This thesis presents a comprehensive numerical study of ground-state boson stars. Boson stars are equilibrium configurations of scalar fields ϕ self-interacting with a quartic repulsive potential, $V(\phi) = \frac{1}{4}\lambda|\phi|^4$. These theoretical, exotic compact objects offer a potential explanation for dark matter, or they could act as black hole mimickers. First, we use a variational approach within General Relativity to derive the Einstein-Klein-Gordon equations that describe the system. Subsequently, we solve the resulting eigenvalue boundary problem numerically with two methods: Python's built-in `solve_bvp` solver and the shooting method.

A comparison of the two numerical approaches demonstrates that, even though their results are in excellent agreement, each method offers unique advantages. While the shooting method guarantees convergence with minimal calibration for single profiles, `solve_bvp` exhibits superior efficiency, executing extensive parameter sweeps over 50 times faster. Physical profiles generated from these solutions reveal that increasing the dimensionless central field, σ_c , produces more compact stars; conversely, increasing the self-interaction strength, Λ , expands the effective extent of the scalar field and supports larger masses. By mapping the relationship between the dimensionless mass, $\mathcal{M}$, and the central field, σ_c , for each configuration family we can determine the characteristic stability boundaries. For non-interacting "mini" boson stars, we evaluate the maximum total mass as $M_{\max} = 0.633 [M_{\text{Pl}}^2/m]$ at a central field of $\sigma_c = 0.271$.

Furthermore, by compiling a dataset of the dimensionless maximum total mass, $\mathcal{M}_{\max}$, and its corresponding self-interaction strength, Λ , we apply linear regression to obtain the empirical expression $\mathcal{M}_{\max} = \sqrt{0.047\Lambda + 0.375}$. This expression accurately reduces to the analytic relation $\mathcal{M}_{\max} \approx 0.22\Lambda^{1/2}$ for large- Λ values ($\Lambda \gg 1$).

Finally, we classify the three evolutionary fates of equilibrium configurations based on their dynamical and energetic stability: stable bound configurations, unstable bound configurations that collapse into black holes, and unstable unbound configurations that disperse away.

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Contents

Περίληψη	1
Abstract	2
Acknowledgments	3
1 Introduction	6
1.1 Maximum mass of "mini" boson stars	7
2 Mathematical Formalism	10
2.1 Conventions	10
2.2 Einstein-Klein-Gordon Equations	10
2.2.1 Klein-Gordon Equation	11
2.2.2 Energy-Momentum Tensor	12
2.2.3 Derivation in Schwarzschild Coordinates	13
2.3 Conserved Quantities	19
3 Numerical Methods	22
3.1 Numerical Approximations	22
3.2 Built-in Python <code>solve_bvp</code> Solver	23
3.2.1 System of First-Order ODEs	24
3.2.2 Boundary Conditions	24
3.2.3 Initial Mesh & Guess	25
3.2.4 Newtonian Limit	26
3.3 Shooting Method	27
3.3.1 Variable Transformation	27
3.3.2 Boundary Conditions	28
3.3.3 Bisection Method	29
3.3.4 Post-Shooting Reconstruction	30
4 Numerical Results: Profiles	31
4.1 Complete Profile of a Boson Star	31
4.2 Parameter Effects on Star Structure	33
4.2.1 Parameter: Central Scalar Field σ_c	33

4.2.2	Parameter: Self-Interaction Strength Λ	35
4.3	Total Mass-Central Field Relationship	36
4.4	Fitting the Maximum Mass— Λ Relation	40
4.5	<code>solve_bvp</code> Solver & Shooting Method Comparison	41
5	Gravitational Stability of Boson Stars	44
5.1	Dynamical & Energetic Stability of Boson Stars	44
5.1.1	Dynamical Stability	44
5.1.2	Energetic Stability	46
5.1.3	The Three Evolutionary Fates of Boson Stars	47
5.1.4	Generalization	49
5.2	Parallels with Neutron Stars: Mass-Radius Relationship	50
6	Applications	54
7	Conclusion	56
	Bibliography	58

Chapter 1

Introduction

Boson stars are stable configurations of massive complex scalar fields [1, 2, 3]. Governed by the laws of general relativity and quantum mechanics [4], these stars are bound together by their own self-gravitation [2], while supported against collapse by quantum mechanical pressure from Heisenberg's uncertainty principle and, when applicable, by a repulsive self-interaction among the particles [3]. This thesis investigates simple boson star models within static [2], asymptotically-flat space-times [1], examining both non-interacting ("mini" boson stars) and self-interacting cases.

Boson stars are of interest for several reasons. Given that axions are among the most compelling dark matter candidates, the detection of axion boson stars could explain dark matter and provide evidence for physics beyond the Standard Model [5]. In addition, galaxy-sized boson stars could form dark matter halos, which would account for the missing cosmological mass and the flat rotation curves of most galaxies [2]. Furthermore, boson stars could act as black hole mimickers with no event horizon and no hard surface, and consequently, binary boson stars could emit gravitational waves with potentially distinguishable signatures [2]. Although the simplified boson star configurations presented in this thesis may not exist in nature, since the self-interaction potential and the resulting equations are relatively simple, they nevertheless provide an excellent model for compact objects across highly variable scales, from particles to stars and galaxies.

The primary focus of this thesis is the use of numerical simulations to determine ground-state configurations of boson stars. To explore more elaborate scenarios than the non-interacting "mini" boson stars, and demonstrate more interesting behaviors, a self-interaction is introduced as an additional support against gravitational collapse. Following Ref. [1], the interaction potential takes the form $V = \frac{1}{4}\lambda|\phi|^4$, where ϕ is the complex scalar field and λ is a dimensionless coupling constant that models the repulsion strength. This formulation is very convenient, since $\lambda = 0$ corresponds to the non-interacting case, and $\lambda > 0$ to the repulsive self-interacting fields. The computational work mainly involves solving the resulting Einstein-Klein-Gordon equations describing the system using two approaches: (i) with the built-in Python `solve_bvp` solver from the `scipy.integrate` package, and (ii) with the shooting method, a standard numerical method for boundary value problems. The scripts and

algorithms developed for this study are publicly available on GitHub at https://github.com/kontheodosiadis/BosonStars_BachelorThesis.

This thesis is organized as follows. The rest of this chapter calculates the maximum mass of a stable "mini" boson star (without self-interaction, $\lambda = 0$) using Heisenberg's uncertainty principle and heuristic, order-of-magnitude scaling arguments. Chapter 2 presents the mathematical formalism for boson stars. This includes the system of Einstein-Klein-Gordon equations describing the spatial evolution of the scalar field and the metric components, restricted to the simple spherically symmetric case with the aforementioned self-interaction potential, and key conserved quantities. Chapter 3 details the numerical methods (Python's `solve_bvp` and the shooting method) used to solve the eigenvalue boundary problem formed by the system of Einstein-Klein-Gordon equations. There, the necessary preparation of the equations and the methodology for each approach are discussed in depth. Special attention is given to the initial guess for each method, as it is the most important factor for ensuring success.

Later chapters are devoted to the physical properties of numerically generated boson star solutions. The scalar field and mass profiles are examined in Chapter 4 for their dependence on the scalar field's central value and the self-interaction strength. Furthermore, from the family of solutions, the evolution of the boson stars' total mass and eigenfrequency as a function of the central value of the scalar field is determined for various self-interaction strengths. These graphs are utilized in Chapter 5 to establish the stability and instability regions of all these boson star configurations. In addition, by compiling a dataset of the maximum stable mass for the various self-interaction strengths, an analytic expression is derived via regression. Finally, Chapter 6 discusses the previously mentioned applications of boson stars in astronomy (sources of dark matter, black hole mimickers and gravitational wave sources).

1.1 Maximum mass of "mini" boson stars

"Mini" boson stars represent the simplest model of boson stars, since they lack self-interaction. In these configurations, the sole outward pressure acting against gravity stems from Heisenberg's uncertainty principle. Therefore, this quantum mechanical pressure dictates the maximum total mass that a "mini" boson star can support before it succumbs to gravitational collapse.

One starts from Heisenberg's position-momentum uncertainty principle of quantum mechanics

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad (1.1)$$

Assuming a constituent particle is confined within a boson star of radius R , its position uncertainty is $\Delta x = R$. Because the particle is effectively bound within a potential well, the momentum expectation value vanishes, $\langle p \rangle = 0$. As a result, the momentum uncertainty simplifies to

$$\Delta p = \sqrt{\langle p^2 \rangle - \cancel{\langle p \rangle^2}^0} = \sqrt{\langle p^2 \rangle} = p$$

Thus, the typical magnitude of the particle's momentum is simply equivalent to its uncertainty, $p = \Delta p$.

Substituting these values into the uncertainty principle (1.1) yields

$$R \cdot p \geq \frac{\hbar}{2} \Rightarrow p \geq \frac{\hbar}{2R} \quad (1.2)$$

This is in agreement with the standard estimate $p \sim \frac{\hbar}{R}$ [1, 2].

One must also assume a moderately relativistic boson star [1], where the maximum momentum of a constituent particle is mc [2], with m being the particle's mass. This imposes an upper bound on the momentum

$$p \leq mc \quad (1.3)$$

Combining these two inequalities for the momentum (1.2) and (1.3) gives

$$\frac{\hbar}{2R} \leq p \leq mc \quad (1.4)$$

Considering the special case where equality holds for both bounds

$$\frac{\hbar}{2R} = mc \Rightarrow R = \frac{\hbar}{2mc} \quad (1.5)$$

This is consistent with the usual requirement for the radius $R \sim \frac{\hbar}{mc}$ [1, 3].

The minimum radius to which an astrophysical object of mass M can be compressed before collapsing into a black hole is the Schwarzschild radius

$$R_s = \frac{2GM}{c^2} \quad (1.6)$$

Since the objective is to find the maximum total mass $M_{\max}$, the heaviest possible star before gravitational collapse occurs, the star's radius approaches its Schwarzschild limit, $R_s = R$. Equating the expressions for the boson star radius R (1.5) and the Schwarzschild radius R_s (1.6), the maximum total mass of the boson star is calculated as

$$M_{\max} = \frac{\hbar c}{4mG} \quad (1.7)$$

Using the definition of the Planck mass, $M_{\text{Pl}} = \sqrt{\hbar c/G}$, this result can be equivalently expressed as

$$M_{\max} = 0.25 \frac{M_{\text{Pl}}^2}{m} \quad (1.8)$$

Equation (1.8) indicates that the maximum mass $M_{\max}$ is on the order of Planck mass squared, divided by the boson mass: M_{Pl}^2/m [6]. This is in complete agreement with the typical scaling $M \sim M_{\text{Pl}}^2/m$ for marginally relativistic boson stars [1, 2, 3].

These simple heuristic calculations demonstrate that the maximum mass of a "mini" boson star is inversely proportional to the mass of the constituent bosons. The wide range of proposed

masses for scalar bosons (spanning from 10^{-22} eV to 10^{13} GeV [7]) subsequently allows a similarly wide range of possible boson star masses. As will be shown in Chapter 2, this inverse proportionality to the mass of the constituent bosons (or equivalently, the proportionality to M_{Pl}^2/m) still holds even when the simple aforementioned self-interaction potential is introduced.

Finally, it is interesting to view boson stars in comparison to fermion stars. On the one hand, in marginally relativistic boson stars, the outward quantum mechanical pressure from the uncertainty principle yields a maximum total mass proportional to M_{Pl}^2/m . On the other hand, in marginally relativistic fermion stars (e.g. neutron stars), the quantum mechanical pressure from Pauli's exclusion principle leads to the characteristic Chandrasekhar mass, which is proportional to M_{Pl}^3/m^2 [1, 8]. In general, the maximum mass for "mini" boson stars is much smaller than that of their fermion counterparts. This comparison is summarized in Table 1.1.

Table 1.1: Comparison between boson stars & fermion stars

	Boson Stars	Fermion Stars
Source of Outward QM Pressure	Heisenberg's Uncertainty Principle	Pauli's Exclusion Principle
Maximum Mass	$M_{\text{max}} \sim M_{\text{Pl}}^2/m$	$M_{\text{ch}} \sim M_{\text{Pl}}^3/m^2$

Chapter 2

Mathematical Formalism

This chapter presents the equations governing boson stars that self-interact under the potential $V(\phi) = \frac{1}{4}\lambda|\phi|^4$. This system of coupled equations comprises the Einstein equations for space-time geometry and the Klein-Gordon equation for the complex scalar field, collectively known as the Einstein-Klein-Gordon (EKG) equations.

In Section 2.2, the EKG equations are derived from a variational approach (using the Lagrangian and the principle of least action) in Schwarzschild coordinates. The key conserved quantities are determined using Noether's theorem in Section 2.3.

2.1 Conventions

Throughout this thesis, Greek letters (α, β, μ, ν , etc.) denote space-time indices ranging from 0 to 3. Furthermore, we adopt natural units, setting $\hbar = c = 1$. In natural units, the Planck mass becomes $M_{Pl} = 1/\sqrt{G}$. Finally, we use the "mostly-plus" signature convention $(-, +, +, +)$ for the metric.

2.2 Einstein-Klein-Gordon Equations

The Einstein-Klein-Gordon equations are obtained via a variational principle. To accommodate both the gravitational field and the other (non-gravitational) fields, the total action S is the sum of the Einstein-Hilbert action S_{EH} for the gravitational field and the "matter" action S_M for any non-gravitational fields present, such as the complex scalar field ϕ [9]

$$S = \frac{1}{16\pi G} S_{EH} + S_M = \int_R \left(\frac{R}{16\pi G} + \mathcal{L}_M \right) \sqrt{|g|} d^4x \quad (2.1)$$

In the equation above, R symbolizes the Ricci scalar, G is the gravitational constant, and $\mathcal{L}_M$ is the Lagrangian for the matter field. The term $d^4V = \sqrt{|g|} d^4x$, where g is the determinant of the metric tensor, corresponds to the invariant volume element for arbitrary coordinates x^μ . In addition, the term $\mathcal{L}\sqrt{|g|}$ is referred to as the Lagrangian density.

The Lagrangian for the complex scalar field ϕ is

$$\mathcal{L} = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi^*\partial_\nu\phi - V(|\phi|^2) = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi^*\partial_\nu\phi - \frac{1}{2}m^2|\phi|^2 - \frac{1}{4}\lambda|\phi|^4 \quad (2.2)$$

The first term can be loosely regarded as the "kinetic energy", while the second and third terms represent the "potential energy" [9]. For the complex scalar field under consideration, the "potential energy" combines the mass term $\frac{1}{2}m^2|\phi|^2$ of the free field (where m is the mass of a constituent particle) and the aforementioned self-interaction $\frac{1}{4}\lambda|\phi|^4$. In spite of the complex scalar field, Ref. [1] includes a factor of 1/2 in the "kinetic energy" simply for later convenience.

2.2.1 Klein-Gordon Equation

To derive the Klein-Gordon equation, one must consider small variations in the complex conjugate of the scalar field ϕ^* of the form $\phi^* \rightarrow \phi^* + \delta\phi^*$, which induce, in turn, variations in the action $S \rightarrow S + \delta S$.

Given that only the matter action depends on the scalar field (and its complex conjugate), the variation is

$$\begin{aligned} \delta S &= \frac{1}{16\pi G} \delta S_{EH} + \delta S_M = \int_R \frac{\delta \mathcal{L}}{\delta \phi^*} \delta \phi^* \sqrt{|g|} d^4x \\ &= \int_R \left\{ \frac{\partial \mathcal{L}}{\partial \phi^*} - \nabla_\mu \left[\frac{\partial \mathcal{L}}{\partial (\nabla_\mu \phi^*)} \right] \right\} \delta \phi^* \sqrt{|g|} d^4x \end{aligned} \quad (2.3)$$

where in the final step, we have substituted the variational derivative $\delta \mathcal{L} / \delta \phi^*$ as defined by Ref. [9].

The principle of least action demands stationarity of the action, $\delta S = 0$. Thus, Ref. [9] obtains the alternative form for the Euler-Lagrange equations

$$\frac{\delta \mathcal{L}}{\delta \phi^*} = \frac{\partial \mathcal{L}}{\partial \phi^*} - \nabla_\mu \left[\frac{\partial \mathcal{L}}{\partial (\nabla_\mu \phi^*)} \right] = 0 \quad (2.4)$$

Because, in this alternative form of the Euler-Lagrange equations, the partial derivative of the Lagrangian is taken with respect to the covariant derivative of the complex conjugate of the field, it is imperative to rewrite the Lagrangian using covariant derivatives. This is easily performed, as the covariant derivative of a scalar is simply the standard partial derivative, $\nabla_\mu \phi = \partial_\mu \phi$. So, (2.2) can be rewritten as

$$\mathcal{L} = -\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi^*\nabla_\nu\phi - \frac{1}{2}m^2\phi\phi^* - \frac{1}{4}\lambda\phi^2(\phi^*)^2 \quad (2.5)$$

For this covariant form of the Lagrangian (2.5), we have

$$\frac{\partial \mathcal{L}}{\partial \phi^*} = -\frac{1}{2}m^2\phi - \frac{1}{2}\lambda\phi^2\phi^* = -\frac{1}{2}m^2\phi - \frac{1}{2}\lambda|\phi|^2\phi \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial (\nabla_\mu \phi^*)} = -\frac{1}{2}g^{\mu\nu}\nabla_\nu\phi$$

Substituting these into the Euler-Lagrange equations (2.4) yields

$$-\frac{1}{2}m^2\phi - \frac{1}{2}\lambda|\phi|^2\phi - \nabla_\mu \left(-\frac{1}{2}g^{\mu\nu}\nabla_\nu\phi \right) = 0$$

Since the covariant derivative of the metric tensor is zero, $\nabla_\mu g^{\mu\nu} = 0$, the field equation satisfied by ϕ is

$$\boxed{\square^2\phi - m^2\phi - \lambda|\phi|^2\phi = 0} \quad (2.6)$$

where $\square^2 \equiv \nabla^\mu \nabla_\mu = g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the covariant d'Alembertian operator.

2.2.2 Energy-Momentum Tensor

In the variational approach to general relativity, the energy-momentum tensor $T_{\mu\nu}$ is defined as

$$T_{\mu\nu} \equiv -\frac{2}{\sqrt{|g|}} \frac{\delta(\mathcal{L}\sqrt{|g|})}{\delta g^{\mu\nu}} \quad (2.7)$$

where $\mathcal{L}\sqrt{|g|}$ is the matter Lagrangian density and $g^{\mu\nu}$ is the inverse metric.

Using this definition, the energy-momentum tensor can be determined by examining the variation of the matter action with respect to the (inverse) metric

$$\delta S_M = \int_R \frac{\delta(\mathcal{L}\sqrt{|g|})}{\delta g^{\mu\nu}} \delta g^{\mu\nu} d^4x = -\frac{1}{2} \int_R T_{\mu\nu} \delta g^{\mu\nu} \sqrt{|g|} d^4x \quad (2.8)$$

Consequently, one must begin by varying the matter action S_M with respect to $g^{\mu\nu}$

$$\delta S_M = \int_R \delta \left(\mathcal{L}\sqrt{|g|} \right) d^4x = \int_R \left(\delta\mathcal{L}\sqrt{|g|} + \mathcal{L}\delta\sqrt{|g|} \right) d^4x \quad (2.9)$$

Because the metric tensor is symmetric, $g^{\mu\nu} = g^{\nu\mu}$, the placement of the indices μ and ν on the partial derivatives of the scalar field and its complex conjugate is interchangeable. Taking advantage of this symmetry, the Lagrangian can be rewritten in a form that does not distinguish between the field and its complex conjugate

$$\mathcal{L} = -\frac{1}{4}g^{\mu\nu} (\partial_\mu\phi^* \partial_\nu\phi + \partial_\nu\phi^* \partial_\mu\phi) - \frac{1}{2}m^2|\phi|^2 - \frac{1}{4}\lambda|\phi|^4 \quad (2.10)$$

Variations in the Lagrangian (2.10) resulting from variations in the inverse metric yield

$$\delta\mathcal{L} = -\frac{1}{4}\delta g^{\mu\nu} (\partial_\mu\phi^* \partial_\nu\phi + \partial_\nu\phi^* \partial_\mu\phi) \quad (2.11)$$

In addition, it can be shown that [9]

$$\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}g_{\mu\nu}\delta g^{\mu\nu} \quad (2.12)$$

Substituting $\delta\mathcal{L}$ and $\delta\sqrt{|g|}$ into (2.9) results in

$$\begin{aligned}\delta S_M &= \int_R \left[-\frac{1}{4} \delta g^{\mu\nu} (\partial_\mu \phi^* \partial_\nu \phi + \partial_\nu \phi^* \partial_\mu \phi) \sqrt{|g|} - \frac{1}{2} \sqrt{|g|} g_{\mu\nu} \delta g^{\mu\nu} \mathcal{L} \right] d^4x \\ &= -\frac{1}{2} \int_R \left[\frac{1}{2} (\partial_\mu \phi^* \partial_\nu \phi + \partial_\nu \phi^* \partial_\mu \phi) + g_{\mu\nu} \mathcal{L} \right] \delta g^{\mu\nu} \sqrt{|g|} d^4x\end{aligned}\quad (2.13)$$

Comparing this result with the general definition of the energy-momentum tensor (2.8) yields

$$\begin{aligned}T_{\mu\nu} &= \frac{1}{2} (\partial_\mu \phi^* \partial_\nu \phi + \partial_\nu \phi^* \partial_\mu \phi) + g_{\mu\nu} \mathcal{L} \\ &= \frac{1}{2} (\partial_\mu \phi^* \partial_\nu \phi + \partial_\nu \phi^* \partial_\mu \phi) - \frac{1}{2} g_{\mu\nu} \left(g^{\alpha\beta} \partial_\alpha \phi^* \partial_\beta \phi + m^2 |\phi|^2 + \frac{1}{2} \lambda |\phi|^4 \right)\end{aligned}\quad (2.14)$$

The Einstein field equations are given by

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (2.15)$$

In this study of the complex scalar field, we are primarily concerned with the 00- and 11-components.

2.2.3 Derivation in Schwarzschild Coordinates

The solutions to Einstein's field equations considered here are spherically symmetric and time-independent, in Schwarzschild coordinates [1]

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -B(r) dt^2 + A(r) dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2) \quad (2.16)$$

The metric tensor forms the diagonal matrix

$$g_{\mu\nu} = \text{diag}(-B(r), A(r), r^2, r^2 \sin^2 \theta)$$

Its determinant is

$$\det(g_{\mu\nu}) = g = -ABr^4 \sin^2 \theta$$

and consequently

$$\sqrt{|g|} = r^2 \sin \theta \sqrt{AB}$$

In addition, the inverse metric tensor is

$$g^{\mu\nu} = \text{diag}(-1/B(r), 1/A(r), 1/r^2, 1/r^2 \sin^2 \theta)$$

Using Schwarzschild coordinates allows us to make the association between the metric com-

ponent $A(r)$ and the mass $M(r)$ enclosed in a sphere of radius r [1, 2]

$$A(r) = \left[1 - \frac{2GM(r)}{r} \right]^{-1} = \left[1 - \frac{2M(r)/M_{\text{Pl}}^2}{r} \right]^{-1} \quad (2.17)$$

We adopt a harmonic ansatz for the complex scalar field [1, 2]

$$\phi(r, t) = \Phi(r)e^{-i\omega t} \quad (2.18)$$

where $\Phi(r) : \mathbb{R} \rightarrow \mathbb{R}$ is a real-valued function.

This harmonic ansatz circumvents Derrick's Theorem, which forbids stable, regular, static field solutions in three-dimensional flat space [2]. Consequently, while the space-time remains static, the scalar field itself is no longer static, but becomes stationary.

The G_{00} and G_{11} components of the Einstein tensor in Schwarzschild coordinates (2.16) are [10]

$$G_{00} = B \left[\frac{1}{rA^2} \frac{dA}{dr} + \frac{1}{r^2} \left(1 - \frac{1}{A} \right) \right] \quad \text{and} \quad G_{11} = A \left[\frac{1}{rAB} \frac{dB}{dr} - \frac{1}{r^2} \left(1 - \frac{1}{A} \right) \right] \quad (2.19)$$

The T_{00} and T_{11} components of the energy-momentum tensor are evaluated from (2.14) using the metric tensor $g_{\mu\nu}$ and the harmonic ansatz (2.18). First, we calculate the term $g^{\alpha\beta} \partial_\alpha \phi^* \partial_\beta \phi$

$$\begin{aligned} g^{\alpha\beta} \partial_\alpha \phi^* \partial_\beta \phi &= g^{00} \partial_0 \phi^* \partial_0 \phi + g^{11} \partial_1 \phi^* \partial_1 \phi + g^{22} \cancel{\partial_2 \phi^* \partial_2 \phi}^0 + g^{33} \cancel{\partial_3 \phi^* \partial_3 \phi}^0 \\ &= -\frac{1}{B} \Phi e^{i\omega t} (+i\omega) \cdot \Phi e^{-i\omega t} (-i\omega) + \frac{1}{A} \frac{d\Phi}{dr} e^{i\omega t} \frac{d\Phi}{dr} e^{-i\omega t} \\ &= -\frac{\omega^2 \Phi^2}{B} + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \end{aligned} \quad (2.20)$$

Here, we have taken advantage of the fact that the metric tensor is diagonal (so only components with $\alpha = \beta$ do not vanish) and that the field $\phi = \phi(r, t)$ depends only on t and r (causing the partial derivatives with respect to φ and θ to vanish).

Taking this into account, the 00-component of the energy-momentum tensor is

$$\begin{aligned} T_{00} &= \frac{1}{2} (\partial_0 \phi^* \partial_0 \phi + \partial_0 \phi \partial_0 \phi^*) - \frac{1}{2} g_{00} \left(g^{\alpha\beta} \partial_\alpha \phi^* \partial_\beta \phi + m^2 |\phi|^2 + \frac{1}{2} \lambda |\phi|^4 \right) \\ &= \Phi e^{i\omega t} (+i\omega) \cdot \Phi e^{-i\omega t} (-i\omega) + \frac{B}{2} \left[-\frac{\omega^2 \Phi^2}{B} + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 + m^2 \Phi^2 + \frac{1}{2} \lambda \Phi^4 \right] \\ &= \frac{B}{2} \left[\left(\frac{\omega^2}{B} + m^2 \right) \Phi^2 + \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right] \end{aligned} \quad (2.21)$$

Similarly, the 11-component of the energy-momentum tensor is

$$\begin{aligned}
T_{11} &= \frac{1}{2} (\partial_1 \phi^* \partial_1 \phi + \partial_1 \phi \partial_1 \phi^*) - \frac{1}{2} g_{11} \left(g^{\alpha\beta} \partial_\alpha \phi^* \partial_\beta \phi + m^2 |\phi|^2 + \frac{1}{2} \lambda |\phi|^4 \right) \\
&= \frac{d\Phi}{dr} e^{i\omega t} \cdot \frac{d\Phi}{dr} e^{-i\omega t} - \frac{1}{2} A \left[-\frac{\omega^2 \Phi^2}{B} + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 + m^2 \Phi^2 + \frac{1}{2} \lambda \Phi^4 \right] \\
&= \frac{A}{2} \left[\left(\frac{\omega^2}{B} - m^2 \right) \Phi^2 - \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right]
\end{aligned} \tag{2.22}$$

For clarity, the 00- and 11- components of the Einstein and the energy-momentum tensors are summarized in Table 2.1

Table 2.1: Tensor Components

Component	Einstein Tensor	Energy-Momentum Tensor
00	$G_{00} = B \left[\frac{1}{rA^2} \frac{dA}{dr} + \frac{1}{r^2} \left(1 - \frac{1}{A} \right) \right]$	$T_{00} = \frac{B}{2} \left[\left(\frac{\omega^2}{B} + m^2 \right) \Phi^2 + \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right]$
11	$G_{11} = A \left[\frac{1}{rAB} \frac{dB}{dr} - \frac{1}{r^2} \left(1 - \frac{1}{A} \right) \right]$	$T_{11} = \frac{A}{2} \left[\left(\frac{\omega^2}{B} - m^2 \right) \Phi^2 - \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right]$

Substituting these expressions into the 00-component of the Einstein field equations (2.15) yields

$$\begin{aligned}
G_{00} &= 8\pi G T_{00} \\
B \left[\frac{1}{rA^2} \frac{dA}{dr} + \frac{1}{r^2} \left(1 - \frac{1}{A} \right) \right] &= 8\pi G \frac{B}{2} \left[\left(\frac{\omega^2}{B} + m^2 \right) \Phi^2 + \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right] \\
\frac{1}{rA^2} \frac{dA}{dr} + \frac{1}{r^2} \left(1 - \frac{1}{A} \right) &= 4\pi G \left[\left(\frac{\omega^2}{B} + m^2 \right) \Phi^2 + \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right]
\end{aligned} \tag{2.23}$$

Similarly, the 11-component of the Einstein field equations (2.15) provides

$$\begin{aligned}
G_{11} &= 8\pi G T_{11} \\
A \left[\frac{1}{rAB} \frac{dB}{dr} - \frac{1}{r^2} \left(1 - \frac{1}{A} \right) \right] &= 8\pi G \frac{A}{2} \left[\left(\frac{\omega^2}{B} - m^2 \right) \Phi^2 - \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right] \\
\frac{1}{rAB} \frac{dB}{dr} - \frac{1}{r^2} \left(1 - \frac{1}{A} \right) &= 4\pi G \left[\left(\frac{\omega^2}{B} - m^2 \right) \Phi^2 - \frac{\lambda}{2} \Phi^4 + \frac{1}{A} \left(\frac{d\Phi}{dr} \right)^2 \right]
\end{aligned} \tag{2.24}$$

With the Einstein equations determined, only the Klein-Gordon equation remains to complete the system of coupled equations.

First, one must evaluate how the covariant d'Alembertian operator acts on the complex scalar field. The operator is expressed as

$$\square^2 \phi \equiv \nabla_\mu \nabla^\mu \phi = \nabla_\mu (g^{\mu\nu} \nabla_\nu \phi)$$

Since the field is scalar, the covariant derivative simplifies to the partial derivative $\nabla_\nu \phi = \partial_\nu \phi$, forming a covariant vector $v_\nu = \partial_\nu \phi$. The expression thus becomes

$$\nabla_\mu (g^{\mu\nu} \partial_\nu \phi) = \nabla_\mu (g^{\mu\nu} v_\nu) = \nabla_\mu v^\mu$$

which is the covariant derivative of a contravariant vector.

By definition, the covariant derivative for a contravariant vector is [10]

$$\nabla_\mu v^\mu = \partial_\mu v^\mu + \Gamma_{\nu\mu}^\mu v^\nu$$

Expressing the Christoffel symbols in terms of the partial derivatives of the metric tensor yields [10]

$$\begin{aligned} \Gamma_{\nu\mu}^\mu &= \frac{1}{2} g^{\alpha\mu} (\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\alpha\mu} - \partial_\alpha g_{\nu\mu}) \\ &= \frac{1}{2} g^{\alpha\mu} \partial_\mu g_{\alpha\nu} + \frac{1}{2} g^{\alpha\mu} \partial_\nu g_{\alpha\mu} - \frac{1}{2} g^{\alpha\mu} \partial_\alpha g_{\nu\mu} \\ &= \frac{1}{2} g^{\alpha\mu} \partial_\nu g_{\alpha\mu} \end{aligned}$$

where, in the final step, the first and third terms cancel out (switching the dummy indices μ and α and utilizing the symmetry of the metric tensor reveals they are identical).

Substituting these results back into the d'Alembertian gives

$$\square^2 \phi = \partial_\mu v^\mu + \frac{1}{2} g^{\alpha\mu} \partial_\nu g_{\alpha\mu} v^\nu = \partial_\mu (g^{\mu\nu} \partial_\nu \phi) + \frac{1}{2} g^{\alpha\mu} g^{\beta\nu} \partial_\beta \phi \partial_\nu g_{\alpha\mu}$$

Evaluating each term separately yields:

For the first term

$$\begin{aligned} \partial_\mu (g^{\mu\nu} \partial_\nu \phi) &= \partial_\mu (g^{\mu 0} \partial_0 \phi + g^{\mu 1} \partial_1 \phi) = \partial_0 (g^{00} \partial_0 \phi) + \partial_1 (g^{11} \partial_1 \phi) \\ &= \partial_0 \left(-\frac{1}{B(r)} \Phi(r) e^{-i\omega t} (-i\omega) \right) + \partial_1 \left(\frac{1}{A(r)} \frac{d\Phi}{dr} e^{-i\omega t} \right) \\ &= -\frac{1}{B(r)} \Phi(r) e^{-i\omega t} (-i\omega)^2 - \frac{1}{A(r)^2} \frac{dA}{dr} \frac{d\Phi}{dr} e^{-i\omega t} + \frac{1}{A(r)} \frac{d^2 \Phi}{dr^2} e^{-i\omega t} \\ &= \left[\frac{\omega^2}{B} \Phi - \frac{1}{A^2} \frac{dA}{dr} \frac{d\Phi}{dr} + \frac{1}{A} \frac{d^2 \Phi}{dr^2} \right] e^{-i\omega t} \end{aligned}$$

For the second term

$$\begin{aligned}
\frac{1}{2}g^{\alpha\mu}g^{\beta\nu}\partial_\beta\phi\partial_\nu g_{\alpha\mu} &= \frac{1}{2}g^{\alpha\mu}g^{0\nu}\partial_0\phi\partial_\nu g_{\alpha\mu} + \frac{1}{2}g^{\alpha\mu}g^{1\nu}\partial_1\phi\partial_\nu g_{\alpha\mu} \\
&= \frac{1}{2}g^{\alpha\mu}g^{00}\partial_0\phi\partial_0 g_{\alpha\mu} + \frac{1}{2}g^{\alpha\mu}g^{11}\partial_1\phi\partial_1 g_{\alpha\mu} \\
&= \frac{1}{2}\frac{1}{A}\frac{d\Phi}{dr}e^{-i\omega t}\left(g^{00}\partial_1 g_{00} + g^{11}\partial_1 g_{11} + g^{22}\partial_1 g_{22} + g^{33}\partial_1 g_{33}\right) \\
&= \frac{1}{2A}\frac{d\Phi}{dr}e^{-i\omega t}\left[-\frac{1}{B}\left(-\frac{dB}{dr}\right) + \frac{1}{A}\frac{dA}{dr} + \frac{1}{r^2}2r\right] + \frac{1}{r^2\sin^2\theta}2r\sin^2\theta \\
&= \frac{1}{2A}\frac{d\Phi}{dr}e^{-i\omega t}\left(\frac{1}{A}\frac{dA}{dr} + \frac{1}{B}\frac{dB}{dr} + \frac{4}{r}\right)
\end{aligned}$$

In both calculations, we have taken into account that the scalar field depends only on radius and time $\phi = \phi(r, t)$, the metric is time-independent with $A = A(r)$, $B = B(r)$, and the metric tensor is diagonal.

Combining these results leads to the final expression for the d'Alembertian operator

$$\square^2\phi = \left[\frac{1}{A}\frac{d^2\Phi}{dr^2} + \frac{1}{A}\left(\frac{2}{r} + \frac{1}{2B}\frac{dB}{dr} - \frac{1}{2A}\frac{dA}{dr}\right)\frac{d\Phi}{dr} + \frac{\omega^2}{B}\Phi\right]e^{-i\omega t} \quad (2.25)$$

Substituting this result into the Klein-Gordon field equation (2.6) yields

$$\begin{aligned}
&\square^2\phi - m^2\phi - \lambda|\phi|^2\phi = 0 \\
&\left[\frac{1}{A}\frac{d^2\Phi}{dr^2} + \frac{1}{A}\left(\frac{2}{r} + \frac{1}{2B}\frac{dB}{dr} - \frac{1}{2A}\frac{dA}{dr}\right)\frac{d\Phi}{dr} + \frac{\omega^2}{B}\Phi\right]e^{-i\omega t} - m^2\Phi e^{-i\omega t} - \lambda\Phi^2\Phi e^{-i\omega t} = 0 \quad (2.26) \\
&\frac{d^2\Phi}{dr^2} + \left(\frac{2}{r} + \frac{1}{2B}\frac{dB}{dr} - \frac{1}{2A}\frac{dA}{dr}\right)\frac{d\Phi}{dr} + \left[\left(\frac{\omega^2}{B} - m^2\right)\Phi - \lambda\Phi^3\right]A = 0
\end{aligned}$$

Equations (2.23), (2.24), (2.26) form the system of coupled Einstein-Klein-Gordon equations. It is standard practice to rescale the variables in such a way that the constants in the equations disappear.

Dividing Equations (2.23) and (2.24) by m^2 gives

$$\begin{aligned}
\frac{1}{m^2rA^2}\frac{dA}{dr} + \frac{1}{(mr)^2}\left(1 - \frac{1}{A}\right) &= \left[\frac{(\omega/m)^2}{B} + 1\right](\sqrt{4\pi G}\Phi)^2 \\
&+ \frac{\lambda}{8\pi Gm^2}(\sqrt{4\pi G}\Phi)^4 + \frac{1}{A}\left[\frac{1}{m}\frac{d(\sqrt{4\pi G}\Phi)}{dr}\right]^2 \quad (2.27)
\end{aligned}$$

$$\begin{aligned}
\frac{1}{m^2rAB}\frac{dB}{dr} - \frac{1}{(mr)^2}\left(1 - \frac{1}{A}\right) &= \left[\frac{(\omega/m)^2}{B} - 1\right](\sqrt{4\pi G}\Phi)^2 \\
&- \frac{\lambda}{8\pi Gm^2}(\sqrt{4\pi G}\Phi)^4 + \frac{1}{A}\left[\frac{1}{m}\frac{d(\sqrt{4\pi G}\Phi)}{dr}\right]^2 \quad (2.28)
\end{aligned}$$

Dividing Equation (2.26) by m^2 and multiplying by $\sqrt{4\pi G}$ yields

$$\begin{aligned} \frac{1}{m^2} \frac{d^2(\sqrt{4\pi G}\Phi)}{dr^2} + \left(\frac{2}{mr} + \frac{1}{2mB} \frac{dB}{dr} - \frac{1}{2mA} \frac{dA}{dr} \right) \frac{1}{m} \frac{d(\sqrt{4\pi G}\Phi)}{dr} \\ + A \left[\left(\frac{(\omega/m)^2}{B} - 1 \right) (\sqrt{4\pi G}\Phi) - \frac{\lambda}{4\pi G m^2} (\sqrt{4\pi G}\Phi)^3 \right] = 0 \end{aligned} \quad (2.29)$$

By introducing dimensionless quantities $x = mr$, $\Omega = \omega/m$, $\sigma = \sqrt{4\pi G}\Phi$ and $\Lambda = \lambda/4\pi G m^2$ and transforming the derivative from r to x with

$$\frac{d}{dr} = \frac{dx}{dr} \frac{d}{dx} = m \frac{d}{dx} \quad \text{and} \quad \frac{d^2}{dr^2} = \frac{d}{dr} \left(m \frac{d}{dx} \right) = m^2 \frac{d^2}{dx^2}$$

the governing equations become

$$\frac{A'}{xA^2} + \frac{1}{x^2} \left(1 - \frac{1}{A} \right) = \left(\frac{\Omega^2}{B} + 1 \right) \sigma^2 + \frac{\Lambda}{2} \sigma^4 + \frac{(\sigma')^2}{A} \quad (2.30a)$$

$$\frac{B'}{xAB} - \frac{1}{x^2} \left(1 - \frac{1}{A} \right) = \left(\frac{\Omega^2}{B} - 1 \right) \sigma^2 - \frac{\Lambda}{2} \sigma^4 + \frac{(\sigma')^2}{A} \quad (2.30b)$$

$$\sigma'' + \left(\frac{2}{x} + \frac{B'}{2B} - \frac{A'}{2A} \right) \sigma' + A \left[\left(\frac{\Omega^2}{B} - 1 \right) \sigma - \Lambda \sigma^3 \right] = 0 \quad (2.30c)$$

Table 2.2: Dimensionality Check

Transformation	(RHS) Units ($c = \hbar = 1$)	(LHS) Units ($c = \hbar = 1$)
$x = mr$	$[m] = \text{GeV}, [r] = \text{GeV}^{-1}$	$[x] = \text{GeV} \cdot \text{GeV}^{-1} = 1$
$\Omega = \omega/m$	$[\omega] = \text{GeV}, [m] = \text{GeV}$	$[\Omega] = \text{GeV}/\text{GeV} = 1$
$\sigma(x) = \sqrt{4\pi G} \Phi(x)$	$[G] = [M_{\text{Pl}}^{-2}] = \text{GeV}^{-2}, [\Phi] = \text{GeV}$	$[\sigma] = \sqrt{\text{GeV}^{-2}} \cdot \text{GeV} = 1$
$\Lambda = \frac{\lambda}{4\pi G m^2}$	$[\lambda] = 1, [G] = \text{GeV}^{-2}, [m] = \text{GeV}$	$[\Lambda] = \frac{1}{\text{GeV}^{-2} \cdot \text{GeV}^2} = 1$

Rescaling the mass expression (2.17) with these dimensionless quantities

$$A(x) = \left[1 - \frac{2M(x)/M_{\text{Pl}}^2}{x/m} \right]^{-1} = \left[1 - \frac{2}{x} \frac{M(x)}{M_{\text{Pl}}^2/m} \right]^{-1} = \left[1 - \frac{2\mathcal{M}(x)}{x} \right]^{-1} \quad (2.31)$$

where we have rescaled the mass into a dimensionless mass

$$\mathcal{M}(x) = \frac{M(x)}{M_{\text{Pl}}^2/m} \quad (2.32)$$

The definition (2.32) confirms the earlier conclusion in Section 1.1 that the mass of boson stars is measured in M_{Pl}^2/m units.

Solving Equation (2.31) gives the dimensionless mass $\mathcal{M}(x)$ enclosed in a sphere of dimen-

sionless radius x

$$\mathcal{M}(x) = \frac{x}{2} \left[1 - \frac{1}{A(x)} \right] \quad (2.33)$$

Equations (2.30a), (2.30b) and (2.30c) are solved subject to the boundary conditions of spatial flatness at the origin $A(0) = 1$ (or equivalently $\mathcal{M}(0) = 0$), regularity at the origin $\sigma(0) = \sigma_c$ and $\sigma'(0) = 0$, and asymptotic flatness at infinity $\sigma(\infty) = 0$ and $B(\infty) = 1$ or $B(\infty) = 1/A(\infty)$ [1, 2, 11]. These equations constitute an eigenvalue problem for Ω subject to the above conditions, concurrently with the restriction that the scalar fields $\Phi(r)$ (or the dimensionless fields $\sigma(x)$) possess no nodes for ground-state solutions.

The solutions are parameterized by the central value of the field σ_c [3]. For a given central value of the field σ_c , we need only to adjust the eigenvalue Ω to find a solution that satisfies all the conditions.

2.3 Conserved Quantities

The Lagrangian (2.2) is invariant under global U(1) transformations [2, 12]

$$\phi \rightarrow \phi' = \phi e^{-i\alpha} \quad (2.34)$$

According to Noether's theorem, this continuous symmetry implies the existence of a conserved current [2, 12]. Because the Lagrangian depends only on the scalar field and its complex conjugate, $\mathcal{L} = \mathcal{L}(\phi, \phi^*, \partial_\mu \phi, \partial_\mu \phi^*)$, the current is defined as

$$J^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \delta \phi^* \quad (2.35)$$

The infinitesimal form of the transformation (2.34) is

$$\begin{aligned} \phi \rightarrow \phi' &= \phi - i\alpha\phi \Rightarrow \delta\phi = \phi' - \phi = -i\alpha\phi \\ \phi^* \rightarrow (\phi^*)' &= \phi^* + i\alpha\phi^* \Rightarrow \delta\phi^* = (\phi^*)' - \phi^* = +i\alpha\phi^* \end{aligned} \quad (2.36)$$

where α is now infinitesimal.

For the specific form of the Lagrangian in (2.2), switching the dummy index μ once in the scalar field $(\partial_\mu \phi)$ and once in the complex conjugate $(\partial_\mu \phi^*)$, we have

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = -\frac{1}{2} g^{\mu\nu} \partial_\nu \phi^* \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} = -\frac{1}{2} g^{\mu\nu} \partial_\nu \phi$$

Substituting these variations and derivatives into the Noether current, and scaling the infinitesimal parameter α out of the current as per standard convention [12], we obtain

$$\alpha J^\mu = -\frac{1}{2} g^{\mu\nu} \partial_\nu \phi^* (-i\alpha\phi) - \frac{1}{2} g^{\mu\nu} \partial_\nu \phi (+i\alpha\phi^*) = \frac{i\alpha}{2} g^{\mu\nu} (\phi \partial_\nu \phi^* - \phi^* \partial_\nu \phi) \quad (2.37)$$

Canceling out α yields the final expression for the Noether Current

$$J^\mu = \frac{i}{2} g^{\mu\nu} (\phi \partial_\nu \phi^* - \phi^* \partial_\nu \phi) \quad (2.38)$$

The associated conserved Noether charge, which corresponds to the total number of boson particles that make up the star [13], is calculated by integrating the time component of the current into the spatial volume [2]

$$N = \int J^0 \sqrt{|g|} d^3x \quad (2.39)$$

Using the harmonic ansatz $\phi(r, t) = \Phi(r) e^{-i\omega t}$ and the diagonal inverse metric with $g^{00} = -1/B(r)$, the time component of the Noether current evaluates to

$$\begin{aligned} J^0 &= \frac{i}{2} g^{0\nu} (\phi \partial_\nu \phi^* - \phi^* \partial_\nu \phi) = \frac{i}{2} g^{00} (\phi \partial_0 \phi^* - \phi^* \partial_0 \phi) \\ &= \frac{i}{2} \left(-\frac{1}{B} \right) [\Phi e^{-i\omega t} \Phi e^{+i\omega t} (+i\omega) - \Phi e^{i\omega t} \Phi e^{-i\omega t} (-i\omega)] = \frac{\omega \Phi^2}{B} \end{aligned} \quad (2.40)$$

Also, recall that the volume element factor is $\sqrt{|g|} = r^2 \sin \theta \sqrt{AB}$.

Given the spherical symmetry of Schwarzschild coordinates (where the spatial coordinates are r, φ, θ), the total conserved particle number is

$$\begin{aligned} N &= \int_{r=0}^{\infty} \int_{\varphi=0}^{2\pi} \int_{\theta=0}^{\pi} \frac{\omega \Phi(r)^2}{B(r)} r^2 \sin \theta \sqrt{A(r)B(r)} dr d\varphi d\theta \\ &= \omega \int_{r=0}^{\infty} r^2 \sqrt{\frac{A(r)}{B(r)}} \Phi(r)^2 dr \int_{\varphi=0}^{2\pi} d\varphi \int_{\theta=0}^{\pi} \sin \theta d\theta \\ &= 4\pi\omega \int_{r=0}^{\infty} r^2 \sqrt{\frac{A(r)}{B(r)}} \Phi(r)^2 dr \end{aligned} \quad (2.41)$$

Finally, we transform this integral using the dimensionless quantities introduced in Section 2.2.3 $x = mr$ (with $dx = mdr$), $\sigma = \sqrt{4\pi G}\Phi$, $\Omega = \omega/m$. This substitution gives

$$\begin{aligned} N &= 4\pi\Omega m \int_{x=0}^{\infty} \frac{x^2}{m^2} \sqrt{\frac{A(x)}{B(x)}} \frac{\sigma(x)^2}{4\pi G} \frac{dx}{m} = \frac{\Omega}{Gm^2} \int_{x=0}^{\infty} x^2 \sqrt{\frac{A(x)}{B(x)}} \sigma(x)^2 dx \\ &= \frac{M_{\text{Pl}}^2}{m^2} \Omega \int_{x=0}^{\infty} x^2 \sqrt{\frac{A(x)}{B(x)}} \sigma(x)^2 dx \end{aligned} \quad (2.42)$$

where, in the final step, we substituted the Planck mass $M_{\text{Pl}} = G^{-1/2}$.

Rescaling the conserved charge to absorb the constants as

$$\mathcal{N} = \frac{N}{(M_{\text{Pl}}^2/m^2)} \quad (2.43)$$

yields the final dimensionless expression for the total particle number

$$\mathcal{N} = \Omega \int_{x=0}^{\infty} x^2 \sqrt{\frac{A(x)}{B(x)}} \sigma(x)^2 dx \quad (2.44)$$

Chapter 3

Numerical Methods

The system of coupled Einstein-Klein-Gordon (EKG) equations is too complex to be solved analytically. This chapter presents two numerical methods used to solve this eigenvalue boundary problem.

Section 3.2 details the first approach, which utilizes Python’s built-in boundary value problem solver (`solve_bvp` from the `scipy.integrate` module). The second approach, the shooting method, which is a standard numerical method for boundary value problems, is described in Section 3.3.

3.1 Numerical Approximations

Transitioning from an analytical description of the physical system to a numerical regime requires various numerical approximations.

To begin with, the independent variable x appears in the denominator of the EKG equations (2.30) and consequently cannot be evaluated directly at $x = 0$. This singularity can be bypassed using a Taylor approximation of the EKG equations around the origin, which would allow the explicit $x = 0$ input [14]. However, for the purposes of this thesis, it suffices to simply approximate zero with a very small lower bound at $\text{eps} = 10^{-5}$. The boundary conditions at the starting point remain unchanged: $\sigma(\text{eps}) = \sigma_c$, $\sigma'(\text{eps}) = 0$, $A(\text{eps}) = 1$.

In addition, the dimensionless radius x for boson stars has no upper limit and can extend to infinity. An elegant way to handle this numerically is to introduce a compactified coordinate $\zeta = \frac{x}{1+x}$ that maps $x \in [0, \infty)$ to $\zeta \in [0, 1]$ [15], and thus eliminates the issues associated with an infinite domain by replacing it with a finite one. However, this compactification is not adopted in this thesis. Instead, for a given central field parameter σ_c , the domain is restricted to a sufficiently large radius x_{max} ($x \in [0, x_{\text{max}}]$), where the scalar field has decayed near zero and the mass has stabilized with high accuracy.

However, replacing infinity with a finite upper bound x_{max} requires adjusting the boundary conditions at the end point. First, rather than enforcing the flat space-time condition $B(\infty) = 1$, setting $B(x_{\text{max}}) = 1/A(x_{\text{max}})$ yields better results, as the metric component A may not have reached its asymptotic value at x_{max} . Furthermore, instead of the Dirichlet boundary

condition $\sigma(\infty) = 0$, the field profile is better determined by the Robin boundary condition $\sigma'(x_{\max}) = -\sqrt{1 - \Omega^2}\sigma(x_{\max})$.

This Robin boundary condition is derived by analyzing the asymptotic behavior of (2.30c) far from the center ($x \gg 1$). One can assume for simplicity that the space-time is approximately Minkowski space-time, such that $A(x) \approx 1$ (with $A'(x) \approx 0$) and $B(x) \approx 1$ (with $B'(x) \approx 0$). Additionally, we assume the dimensionless field has decayed to a very small amplitude ($\sigma \ll 1$ and $\sigma' \ll 1$), so that all non-linear terms can be neglected. Under these assumptions, (2.30c) simplifies to

$$\sigma'' + (\Omega^2 - 1)\sigma = 0 \quad (3.1)$$

For bound boson star configurations, the scalar field must decay asymptotically and satisfy the boundary condition $\sigma(\infty) = 0$. The general solution to this differential equation consists of real exponentials

$$\sigma(x) = C_1 \exp\left(\sqrt{1 - \Omega^2}x\right) + C_2 \exp\left(-\sqrt{1 - \Omega^2}x\right) \quad (3.2)$$

provided that $1 - \Omega^2 > 0$, implying the restriction $\Omega < 1$. For $\Omega > 1$, the general solutions become oscillatory (sines and cosines). These oscillatory solutions do not decay and cannot satisfy the asymptotic boundary condition $\sigma(\infty) = 0$, as such a limit does not mathematically exist for these functions. Therefore, bound boson star configurations must strictly satisfy the constraint $\omega/m = \Omega \leq 1$ [16].

To satisfy the boundary condition $\sigma(\infty) = 0$, it is necessary to set $C_1 = 0$ to prevent the solution from diverging to infinity. Therefore, the physically acceptable solution takes the form:

$$\sigma(x) = C_2 \exp\left(-\sqrt{1 - \Omega^2}x\right) \quad (3.3)$$

The constant C_2 can be eliminated by differentiating this solution

$$\sigma'(x) = C_2 \exp\left(-\sqrt{1 - \Omega^2}x\right) \cdot \left(-\sqrt{1 - \Omega^2}\right) = -\sqrt{1 - \Omega^2}\sigma(x) \quad (3.4)$$

Evaluating (3.4) at the numerical boundary $x_{\max}$ yields the stated Robin condition.

Finally, all the configurations in this thesis are modeled using the fully relativistic EKG system (2.30). The Newtonian limit can be used for amplitudes $\sigma \leq 10^{-3}$ where the EKG system is reduced to the Schrödinger-Newton (or Schrödinger-Poisson) system [16, 2]. Nonetheless, the configurations are always derived from the EKG system.

3.2 Built-in Python solve_bvp Solver

The `solve_bvp` function, available within Python's `scipy.integrate` module, numerically solves a system of first-order ordinary differential equations (ODEs) subject to two-point boundary conditions [17]. The solver "implements a fourth-order collocation algorithm with the control of residuals" [17], where the non-linear "collocation system is solved by a damped Newton

method" [17].

The `solve_bvp` function takes an initial guess for both the dependent variables at the specified mesh nodes and any unknown parameters/eigenvalues. It then returns the evaluated solution at these nodes along with the found parameters/eigenvalues [17]. The convergence and the overall success of the solver are highly sensitive to the quality of this initial guess.

Furthermore, ensuring that the problem is well-posed requires specifying the exact number of boundary conditions mandated by the system's degrees of freedom. For a system of first-order ODEs with n dependent variables and k parameters which are to be found along with them, there must be exactly $n + k$ boundary conditions for the problem to be determined [17].

3.2.1 System of First-Order ODEs

To utilize `solve_bvp`, the coupled differential equations must be written as a system of first-order ODEs. A system of n first-order equations is defined as [18]

$$\begin{aligned}\frac{dy_1}{dt} &= f_1(y_1, y_2, \dots, y_n, t) \\ \frac{dy_2}{dt} &= f_2(y_1, y_2, \dots, y_n, t) \\ &\vdots \\ \frac{dy_n}{dt} &= f_n(y_1, y_2, \dots, y_n, t)\end{aligned}\tag{3.5}$$

The EKG equations (2.30) cannot be directly written in this form because Equation (2.30c) contains a second-order derivative of the scalar field σ'' . Following the standard procedure for reducing a second-order equation to a set of first-order ODEs, we introduce an auxiliary variable $v(x) \equiv \sigma'(x)$ with $v'(x) = \sigma''(x)$ [19]. Consequently, the EKG system can be written as the following first-order system

$$A' = xA^2 \left[\left(\frac{\Omega^2}{B} + 1 \right) \sigma^2 + \frac{\Lambda}{2} \sigma^4 + \frac{v^2}{A} \right] - \frac{A}{x} (A - 1) = f_1(A, B, \sigma, v, x) \tag{3.6a}$$

$$B' = xAB \left[\left(\frac{\Omega^2}{B} - 1 \right) \sigma^2 - \frac{\Lambda}{2} \sigma^4 + \frac{v^2}{A} \right] + \frac{B}{x} (A - 1) = f_2(A, B, \sigma, v, x) \tag{3.6b}$$

$$\sigma' = v = f_3(A, B, \sigma, v, x) \tag{3.6c}$$

$$v' = - \left(\frac{2}{x} + \frac{B'}{2B} - \frac{A'}{2A} \right) v - A \left[\left(\frac{\Omega^2}{B} - 1 \right) \sigma - \Lambda \sigma^3 \right] = f_4(A, B, \sigma, v, x) \tag{3.6d}$$

where A' and B' in the expression of v' are implicitly substituted by the functions $f_1(A, B, \sigma, v, x)$ and $f_2(A, B, \sigma, v, x)$, respectively.

3.2.2 Boundary Conditions

As established previously, a system with n dependent variables and k unknown parameters requires $n + k$ boundary conditions to be well-posed. The first-order EKG system contains

$n = 4$ dependent variables (A, B, σ, v) and $k = 1$ unknown parameter (the eigenvalue Ω). Therefore, the problem requires 5 boundary conditions.

The enforced boundary conditions are: $A(\text{eps}) = 1$, $\sigma(\text{eps}) = \sigma_c$, $v(\text{eps}) = 0$ at the inner boundary, along with $B(x_{\text{max}}) = 1/A(x_{\text{max}})$ and $\sigma'(x_{\text{max}}) = -\sqrt{1 - \Omega^2}\sigma(x_{\text{max}})$ at the outer boundary. This satisfies the requirement of 5 boundary conditions, ensuring the boundary value problem is well-posed. Note that the physical requirement of no nodes is not a boundary condition itself, but rather a constraint to confine the results to ground-state configurations.

3.2.3 Initial Mesh & Guess

The mesh must range from the starting point at $x = \text{eps}$ to the ending point at $x = x_{\text{max}}$. The initial mesh nodes must be chosen carefully. On the one hand, an excessively dense initial mesh undermines the efficiency of the adaptive mesh refinement implemented by `solve_bvp` [17], resulting in a massive collocation system that often triggers a *max_nodes exceeded* error. Conversely, an overly coarse initial mesh fails to provide adequate resolution for the Newton solver, causing the problem to diverge and lead to a *Singular Jacobian* error.

Crucial to convergence is the initial guess for the functions and the eigenvalue. Because of the Newton-like method inside the `solve_bvp` solver, a poor or unrealistic initial guess will either converge to an unintended branch (such as an excited state with nodes) or cause the solver to diverge entirely.

After various attempts, the initial guesses were chosen as follows:

For the metric components $A(x)$ and $B(x)$ we initialize in a flat Minkowski space-time

$$A(x) = 1 \quad \text{and} \quad B(x) = 1 \quad (3.7)$$

By starting from a gravity-free space-time, the solver makes the necessary adjustments due to gravity as it iterates towards the solution.

For the scalar field $\sigma(x)$ the trial function is a Gaussian

$$\sigma(x) = \sigma_c \exp\left(-\frac{x^2}{w^2}\right) \quad (3.8)$$

where the characteristic width is $w = \max(8, 1/\sigma_c)$ based on considerations from the Newtonian limit. This Gaussian function fulfills the criteria for a nodeless ground-state by beginning at the central value σ_c and decaying asymptotically toward zero without crossing the horizontal axis, as illustrated in Figure 3.1.

For the auxiliary variable $v(x)$ rather than supplying an arbitrary guess, we differentiate the scalar field guess (3.8)

$$v(x) = \sigma'(x) = -2\sigma_c \frac{x}{w^2} \exp\left(-\frac{x^2}{w^2}\right) \quad (3.9)$$

which inherently satisfies the condition $\sigma'(0) = 0$ at the origin. This highlights the advantage

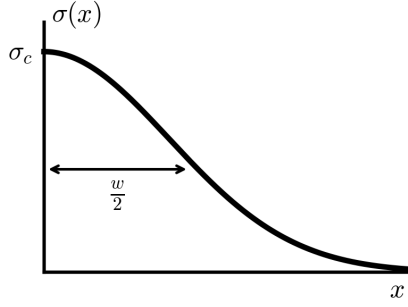


Figure 3.1: Gaussian trial function

of using a Gaussian trial function over a simple exponential.

For the eigenvalue Ω the initial guess is motivated by the Newtonian limit

$$\Omega = \sqrt{1 - \sigma_c^2} \quad (3.10)$$

This choice enforces $\Omega \leq 1$, while capturing the behavior $\Omega \approx 1$ as $\sigma_c \leq 10^{-3}$ [16, 2].

For sequential parameter sweeps, once a configuration has been determined for a given central field value σ_c , its solutions can be used as a much better initial guess for the next incremental value, $\sigma_{c(\text{new})}$.

3.2.4 Newtonian Limit

In the Newtonian limit, the EKG system of equations is reduced to the Schrödinger-Newton system of equations [16]. Under this weak field approximation, fields are of order $O(\epsilon^2) \ll 1$ meaning only leading-order terms are preserved. Consequently, the fourth-order self-interaction term vanishes.

The Schrödinger-Newton system exhibits a scaling symmetry: if $\{\tau, x, U, \sigma\}$ comprises a solution, then $\{\hat{\tau}/k^2, \hat{x}/k, k^2\hat{U}, k^2\hat{\sigma}\}$ is also a solution [20, 16], where $\tau = mt$ is the dimensionless time, $x = mr$ is the dimensionless radius, U is the Newtonian gravitational potential, σ is the dimensionless field and k is an arbitrary non-zero scaling parameter.

Analytically, the dimensionless frequency satisfies $\Omega^2 = 1 - k^2$ [16]. The weak field restriction implies $k^2 \leq 10^{-3}$ [16], which yields $\Omega \approx 1$. From the scaling properties it follows that $\sigma(x) = k^2\hat{\sigma}(x)$. Applying this relation at the origin $\sigma_c = k^2\hat{\sigma}_c$ and adopting the standard normalization $\hat{\sigma}_c = 1$ [16] we obtain $\sigma_c = k^2$. This yields $\Omega = \sqrt{1 - \sigma_c}$.

Because the initial central field will not exceed 1, $\sigma_c \leq 1$, it follows $\sigma_c^2 \leq \sigma_c$. As a failsafe against divergence in the first internal iteration, we intentionally initialize Ω closer to 1 than the Newtonian limit prediction. Therefore, the analytic relation is modified to $\Omega = \sqrt{1 - \sigma_c^2}$.

In addition, the scaling property implies that $x \propto 1/k$. Given that $\sigma_c = k^2$, it scales as $x \propto 1/\sqrt{\sigma_c}$. Because $\sigma_c \leq \sqrt{\sigma_c}$ for values below 1, it follows $1/\sigma_c \geq 1/\sqrt{\sigma_c}$. We intentionally inflate the width of the Gaussian trial function $w \propto 1/\sigma_c$ to guarantee safe convergence. However, this is not enough. Finally, imposing a hard minimum bound $\text{max}(8, \dots)$ prevents the highly relativistic configurations (large central fields σ_c) from shrinking into a sharp spike,

for which the solver runs a high risk of diverging.

3.3 Shooting Method

The shooting method was developed with the goal of transforming an ODE boundary value problem into an equivalent initial value problem [21]. This transformation is advantageous because integrating an ODE initial value problem is mathematically straightforward: one simply starts from the initial value and advances the system forward to obtain the solution [21]. However, a boundary value problem inherently does not have enough initial conditions at a single boundary, and as a result cannot yield a unique solution [19].

In the shooting method, we choose values for all the dependent variables at one boundary, supplying initial guesses for the ones missing [19]. Then, the resulting ODE initial value problem is integrated to the other boundary, where the outcome is compared with the desired boundary values [21]. Based on the discrepancies, the initial guess is adjusted and the process is repeated until the error is acceptable [21]. Finding the optimal guess that nullifies the discrepancies at the other boundary constitutes a root-finding problem [19].

For the EKG coupled equations, the initial value problem is integrated using the Runge-Kutta-Fehlberg method (RK45). Furthermore, the bisection method was selected for the root-finding problem, because it is the easiest to implement and cannot fail [19].

3.3.1 Variable Transformation

The EKG eigenvalue boundary problem, as a first-order system of ODEs, consists of 4 dependent variables (A , B , σ , $v = \sigma'$) and 1 eigenvalue (Ω). The boundary conditions are distributed with 3 at the starting point ($A(0) = 1$, $\sigma(0) = \sigma_c$, $v(0) = \sigma'(0) = 0$, with their numerical equivalents) and the remaining 2 at the end point ($B(\infty) = 1$, $\sigma(\infty) = 0$, with their numerical equivalents). Consequently, a direct implementation of the shooting method would require 2 shooting parameters: the missing dependent value from the starting point $B(0)$ and the eigenvalue Ω . This results in a multi-dimensional root-finding problem, making its solution both complex and computationally expensive. Fortunately, through mathematical transformations, the problem can be reduced to a single parameter shooting problem, which is significantly more efficient to solve.

The transformation focuses on the metric component $B(x)$, as it lacks a known condition at the origin. Introducing the rescaled metric component [3]

$$\tilde{B}(x) = \frac{B(x)}{B(0)} \quad (3.11)$$

provides an elegant mechanism to move the boundary condition from the end point back to the starting point. Under this definition, the condition at the origin becomes $\tilde{B}(0) = B(0)/B(0) = 1$, leaving the value at infinity undetermined $\tilde{B}(\infty) = B(\infty)/B(0) = 1/B(0)$.

Applying the transformation (3.11) $B(x) = B(0)\tilde{B}(x)$ and $B'(x) = B(0)\tilde{B}'(x)$ on the EKG system of equations (2.30) gives

$$\frac{A'}{xA^2} + \frac{1}{x^2} \left(1 - \frac{1}{A}\right) = \left(\frac{\Omega^2}{B(0)\tilde{B}} + 1\right) \sigma^2 + \frac{\Lambda}{2}\sigma^4 + \frac{(\sigma')^2}{A} \quad (3.12a)$$

$$\frac{\tilde{B}'}{xA\tilde{B}} - \frac{1}{x^2} \left(1 - \frac{1}{A}\right) = \left(\frac{\Omega^2}{B(0)\tilde{B}} - 1\right) \sigma^2 - \frac{\Lambda}{2}\sigma^4 + \frac{(\sigma')^2}{A} \quad (3.12b)$$

$$\sigma'' + \left(\frac{2}{x} + \frac{\tilde{B}'}{2A\tilde{B}} - \frac{A'}{2A}\right) \sigma' + A \left[\left(\frac{\Omega^2}{B(0)\tilde{B}} - 1\right) \sigma - \Lambda\sigma^3 \right] = 0 \quad (3.12c)$$

With the intention of writing the transformed EKG system in the same form as the original, we introduce a rescaled eigenvalue

$$\tilde{\Omega}^2 = \frac{\Omega^2}{B(0)} \quad \text{or} \quad \tilde{\Omega} = \frac{\Omega}{\sqrt{B(0)}} \quad (3.13)$$

The final simplified system is

$$\frac{A'}{xA^2} + \frac{1}{x^2} \left(1 - \frac{1}{A}\right) = \left(\frac{\tilde{\Omega}^2}{\tilde{B}} + 1\right) \sigma^2 + \frac{\Lambda}{2}\sigma^4 + \frac{(\sigma')^2}{A} \quad (3.14a)$$

$$\frac{\tilde{B}'}{xA\tilde{B}} - \frac{1}{x^2} \left(1 - \frac{1}{A}\right) = \left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1\right) \sigma^2 - \frac{\Lambda}{2}\sigma^4 + \frac{(\sigma')^2}{A} \quad (3.14b)$$

$$\sigma'' + \left(\frac{2}{x} + \frac{\tilde{B}'}{2A\tilde{B}} - \frac{A'}{2A}\right) \sigma' + A \left[\left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1\right) \sigma - \Lambda\sigma^3 \right] = 0 \quad (3.14c)$$

As detailed in Section 3.2, the EKG system must be written as a set of first-order ODEs. For this purpose, an auxiliary variable $v(x) = \sigma'(x)$ is introduced to eliminate the second-order derivative of the scalar field. The resulting system of first-order ODEs assumes the same form as before in (3.6), but parametrized by $\tilde{B}$ and $\tilde{\Omega}$ instead of B and Ω

$$A' = xA^2 \left[\left(\frac{\tilde{\Omega}^2}{\tilde{B}} + 1\right) \sigma^2 + \frac{\Lambda}{2}\sigma^4 + \frac{v^2}{A} \right] - \frac{A}{x}(A - 1) = f_1(A, \tilde{B}, \sigma, v, x) \quad (3.15a)$$

$$\tilde{B}' = xA\tilde{B} \left[\left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1\right) \sigma^2 - \frac{\Lambda}{2}\sigma^4 + \frac{v^2}{A} \right] + \frac{\tilde{B}}{x}(A - 1) = f_2(A, \tilde{B}, \sigma, v, x) \quad (3.15b)$$

$$\sigma' = v = f_3(A, \tilde{B}, \sigma, v, x) \quad (3.15c)$$

$$v' = - \left(\frac{2}{x} + \frac{\tilde{B}'}{2A\tilde{B}} - \frac{A'}{2A}\right) v - A \left[\left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1\right) \sigma - \Lambda\sigma^3 \right] = f_4(A, \tilde{B}, \sigma, v, x) \quad (3.15d)$$

3.3.2 Boundary Conditions

Following the variable transformation $B \rightarrow \tilde{B}$ the enforced boundary conditions are: $A(0) = 1$, $\tilde{B}(0) = 1$, $\sigma(0) = \sigma_c$, $v(0) = \sigma'(0) = 0$ and $\sigma(\infty) = 0$ (implemented using the numerical approximations discussed in Section 3.1). Furthermore, there is the additional constraint that

the scalar field amplitude must exhibit zero nodes.

The adjusted EKG system still consists of 4 dependent variables (A , $\tilde{B}$, σ , $v = \sigma'$) and 1 eigenvalue ($\tilde{\Omega}$). However, the boundary conditions for this problem are now distributed with 4 at the starting point ($A(0) = 1$, $\tilde{B}(0) = 1$, $\sigma(0) = \sigma_c$, $v(0) = \sigma'(0) = 0$, with their numerical equivalents) and the remaining 1 at the end point ($\sigma(\infty) = 0$, with its numerical equivalent). This leaves $\tilde{\Omega}$ as the sole shooting parameter, whose precise value is iteratively determined via the Bisection Method.

3.3.3 Bisection Method

The bisection method relies on the intermediate value theorem, which states that if a function is continuous in an interval and has opposite values at its endpoints, then the root is bracketed in that interval (at least one root must lie in that interval) [19]. Determining the eigenvalue $\tilde{\Omega}$ is not explicitly formulated as a standard zero-crossing problem, therefore there is no direct sign change to evaluate. Instead, the method is adjusted to bracket the true eigenvalue $\tilde{\Omega}$ between a lower $\tilde{\Omega}_{\text{low}}$ and upper $\tilde{\Omega}_{\text{high}}$ bound, where the classification of a value as either "low" or "high" is triggered by a different event while integrating the ODE initial value problem, as discussed below.

The bisection algorithm computes the interval midpoint $\tilde{\Omega}_m = \frac{\tilde{\Omega}_{\text{low}} + \tilde{\Omega}_{\text{high}}}{2}$, evaluates its low/high state, and replaces whichever limit shares the same state [19]. Specifically, there are 3 scenarios: (i) if the integration reaches the end point $x = x_{\text{max}}$ without triggering any events, the midpoint represents the true eigenvalue (or a value within acceptable tolerance); (ii) if the midpoint triggers the event associated with a value lower than the eigenvalue, then $\tilde{\Omega}_m$ improves the lower bound by replacing $\tilde{\Omega}_{\text{low}}$; and (iii) if the midpoint triggers the event associated with a value higher than the eigenvalue, then $\tilde{\Omega}_m$ improves the upper bound by replacing $\tilde{\Omega}_{\text{high}}$ [21]. This process is repeated, and after each iteration the bounds containing the actual eigenvalue decrease by a factor of two.

The two events that determine whether a value for $\tilde{\Omega}$ is lower or higher than the actual eigenvalue are: divergence to infinity (violating $\sigma(\infty) = 0$) and the appearance of a node (violating the 0-nodes constraint), respectively.

This behavior is understood by examining (3.14c) through a mechanical spring analogue. For simplicity, if the self-interaction is ignored ($\Lambda = 0$), the equation reads

$$\begin{aligned} \sigma'' + \left(\frac{2}{x} + \frac{\tilde{B}'}{2\tilde{B}} - \frac{A'}{2A} \right) \sigma' + A \left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1 \right) \sigma &= 0 \\ \sigma'' + \text{friction} \cdot \sigma' + A \left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1 \right) \sigma &= 0 \end{aligned} \tag{3.16}$$

This analysis is most valid near the center, where the condition $\tilde{\Omega}^2/\tilde{B} \geq 1$ holds to keep the star bound. Under this consideration, Equation (3.16) mimics a damped harmonic oscillator, where the term $A \left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1 \right)$ acts as an effective spring constant. If $\tilde{\Omega}$ is too high, it behaves

like an overly stiff spring: the restoring force is excessively strong, pulling the scalar field amplitude σ down so fast that it overshoots the axis, creating a node. Thus, the detection of a node corresponds to a high value for $\tilde{\Omega}$. If $\tilde{\Omega}$ is too low, it acts like a weak spring: the restoring force is insufficient to pull σ toward zero. As a result, the field bottoms out before reaching asymptotic infinity, and the non-linear coupling eventually drives the solution to diverge and shoot off toward infinity. Therefore, divergence and the violation of the condition $\sigma(\infty) = 0$ is associated with a low value for $\tilde{\Omega}$.

Interestingly, when the self-interaction term is included ($\Lambda \neq 0$), the cubic non-linear stiffness transforms the equation into a damped Duffing oscillator [22, 23].

$$\sigma'' + \text{friction} \cdot \sigma' + A \left(\frac{\tilde{\Omega}^2}{\tilde{B}} - 1 \right) \sigma - A\Lambda\sigma^3 = 0 \quad (3.17)$$

3.3.4 Post-Shooting Reconstruction

The completion of the shooting method yields results for the shooting parameter $\tilde{\Omega}$ and the rescaled metric component $\tilde{B}(x)$. However, the ultimate objective is to evaluate the true Ω and $B(x)$. This inverse transformation requires calculating the unknown scaling factor $B(0)$. Since the only known value for $B(x)$ is defined at the end point $x = x_{\max}$, $B(x_{\max}) = 1/A(x_{\max})$, it is extracted

$$\tilde{B}(x_{\max}) = \frac{B(x_{\max})}{B(0)} \Rightarrow B(0) = \frac{1}{A(x_{\max})\tilde{B}(x_{\max})} \quad (3.18)$$

With $B(0)$ determined, via direct substitution the true variables are

$$B(x) = B(0)\tilde{B}(x) = \frac{\tilde{B}(x)}{A(x_{\max})\tilde{B}(x_{\max})} \quad (3.19)$$

and

$$\Omega = \tilde{\Omega}\sqrt{B(0)} = \frac{\tilde{\Omega}}{\sqrt{A(x_{\max})\tilde{B}(x_{\max})}} \quad (3.20)$$

Chapter 4

Numerical Results: Profiles

This chapter presents and analyzes the numerical results obtained from solving the Einstein-Klein-Gordon (EKG) equations with the methods discussed in Chapter 3. These results are primarily visualized through diagrams and supported by relevant data tables.

The chapter begins in Section 4.1 by presenting the complete profile of a boson star—specifically, its metric components, dimensionless scalar field and dimensionless mass as a function of radius. Section 4.2 investigates how the star’s structure is influenced by the two model parameters: the central scalar field, σ_c , and the self-interaction strength, Λ . Subsequent sections focus more narrowly on the effect of these parameters on the dimensionless total mass of the star. In particular, Section 4.3 examines the relationship between the total mass and the central field, while Section 4.4 fits the maximum total mass and self-interaction strength data points. Finally, the chapter concludes in Section 4.5 with a preliminary comparison between Python’s built-in `solve_bvp` solver and the shooting method in terms of execution time.

4.1 Complete Profile of a Boson Star

A solution of the EKG system explicitly yields the metric components, $A(x)$ and $B(x)$, and the dimensionless scalar field, $\sigma(x)$, as functions of the dimensionless radius x , alongside the eigenfrequency Ω . These results are subsequently used in Equation (2.33) to determine the dimensionless mass distribution inside a sphere of radius x , and in Equation (2.44) to calculate the total number of particles constituting the boson star. As discussed in Section 2.2.3, the solution is parameterized solely by the central scalar field, σ_c , for a given self-interaction strength, Λ .

The following results pertain to a boson star where $\sigma_c = 0.1$ and $\Lambda = 0$. Figure 4.1 illustrates the complete profile of this boson star ($\sigma_c = 0.1$, $\Lambda = 0$): its metric components, scalar field and mass plotted against the dimensionless radius x . This visual profile is complemented by Table 4.1 that details the star’s macroscopic properties: its eigenfrequency Ω , its total mass M (in M_{Pl}^2/m units) and the total number of constituent particles N (in M_{Pl}^2/m^2 units). Although enforcing a zero node constraint makes the node count redundant, it is included in the table for completeness. In addition to these macroscopic properties, the table provides the boundary

values that were either approximated implicitly (see Section 3.1) or previously unknown.

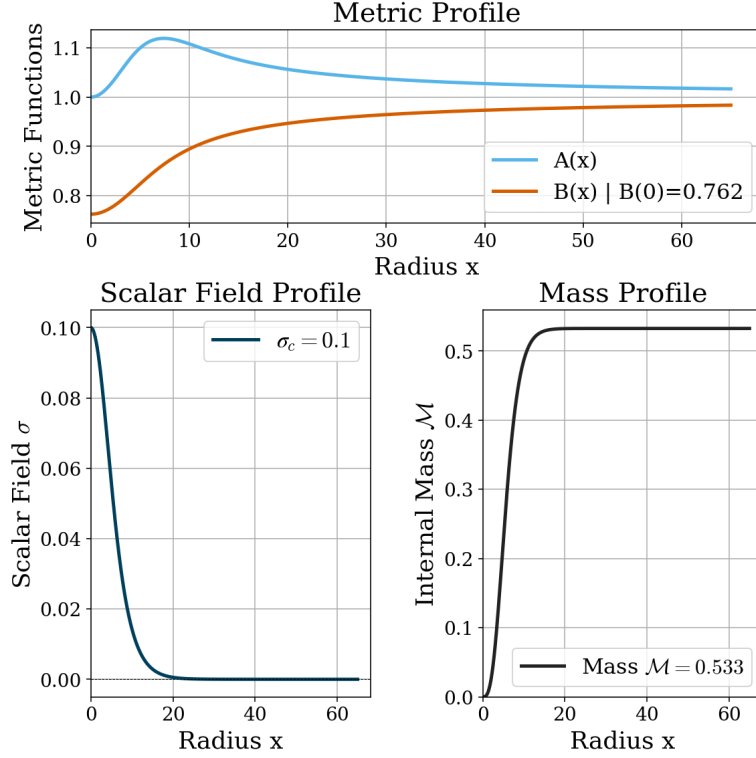


Figure 4.1: Boson star profile: Metric, dimensionless scalar field & dimensionless mass ($\sigma_c = 0.1$ and $\Lambda = 0$)

A qualitative analysis of the diagrams reveals the behavioral trends. The scalar field begins at its central value of $\sigma_c = 0.1$ and decreases monotonically, eventually flattening to approach zero asymptotically. The field never overshoots the axis creating a node, complying with the zero node constraint. In a similar way, the mass distribution starts at zero at the star's center and rises monotonically until it stabilizes, asymptotically approaching the star's dimensionless total mass $\mathcal{M}$.

The behavior of the metric components near the outer boundary is of particular interest. The graph clearly shows that $A(x)$ has not reached its asymptotic value of 1 at $x = x_{\max}$. Therefore, enforcing the boundary condition $B(x_{\max}) = 1$ would introduce significant inaccuracy to the solution. This observation justifies choosing $B(x_{\max}) = 1/A(x_{\max})$ to improve the solution's precision.

These qualitative observations are verified by a quantitative analysis. The boundary value of $A(x_{\max}) = 1.017 \neq 1$ confirms that the metric component has not decayed to its asymptotic limit. Conversely, the scalar field at the boundary $x = x_{\max}$ is infinitesimally small, having decayed to the order of 10^{-10} .

Moreover, to further support the asymptotic approach of the scalar field towards zero and the stabilization of the mass to its total value $\mathcal{M}$, Table 4.2 presents their final five values generated by the solver. Within that interval, the scalar field consistently remains close to zero, maintaining the order 10^{-10} . Meanwhile, the mass values remain exactly the same with an accuracy of four decimals, and the percentage difference from the final value shows only a

Table 4.1: Summary of profile results

Macroscopic Properties			
Nodes	Eigenfrequency Ω	Mass M [M_{Pl}^2/m]	Particles N [M_{Pl}^2/m^2]
0	0.937	0.533	0.543

Boundary Values			
$\sigma(x_{\text{max}})$	$A(x_{\text{max}})$	$B(x_{\text{max}})$	$B(\text{eps})$
1.016×10^{-10}	1.017	0.984	0.762

minuscule divergence on the order of 10^{-12} or smaller.

Table 4.2: Comparison of the last 5 values for radius, scalar field, and mass

Index	-5	-4	-3	-2	-1
Radius	64.740	64.805	64.870	64.935	65.000
Scalar Field					
Value	1.112×10^{-10}	1.087×10^{-10}	1.063×10^{-10}	1.039×10^{-10}	1.016×10^{-10}
% Diff	9.538%	7.072%	4.662%	2.305%	
Mass					
Value	0.5326	0.5326	0.5326	0.5326	0.5326
% Diff	$-2.710 \times 10^{-13}\%$	$-1.147 \times 10^{-12}\%$	$-1.063 \times 10^{-12}\%$	$-5.003 \times 10^{-13}\%$	

4.2 Parameter Effects on Star Structure

The boson star model is characterized solely by two parameters: the central field density, σ_c , and the self-interaction strength, Λ . To isolate the impact of each parameter on the stellar structure, they are examined individually by varying one value while holding the other constant. Rather than displaying an exhaustive profile for every configuration, the investigation is focused on the radial distributions of the scalar field and mass, alongside the star’s macroscopic properties.

4.2.1 Parameter: Central Scalar Field σ_c

The influence of the central field parameter, σ_c , is analyzed by varying its values while keeping the self-interaction strength constant ($\Lambda = \text{const.}$). The following results pertain to boson stars with $\Lambda = 0$. Figure 4.2 illustrates the scalar field and mass profiles for a range of central field values. These diagrams are supplemented by Table 4.3, which details the corresponding macroscopic properties for each configuration: the eigenfrequency Ω , the total mass M (in M_{Pl}^2/m units) and the total number of particles constituting the stars N (in M_{Pl}^2/m^2 units). Because the boundary values remain very close to those documented in Table 4.1—with $A(x_{\text{max}})$ and $B(x_{\text{max}})$ approaching their asymptotic limit of 1 even more closely, and $\sigma(x_{\text{max}})$ remaining at or below the order of 10^{-7} —they are omitted from this table for brevity.

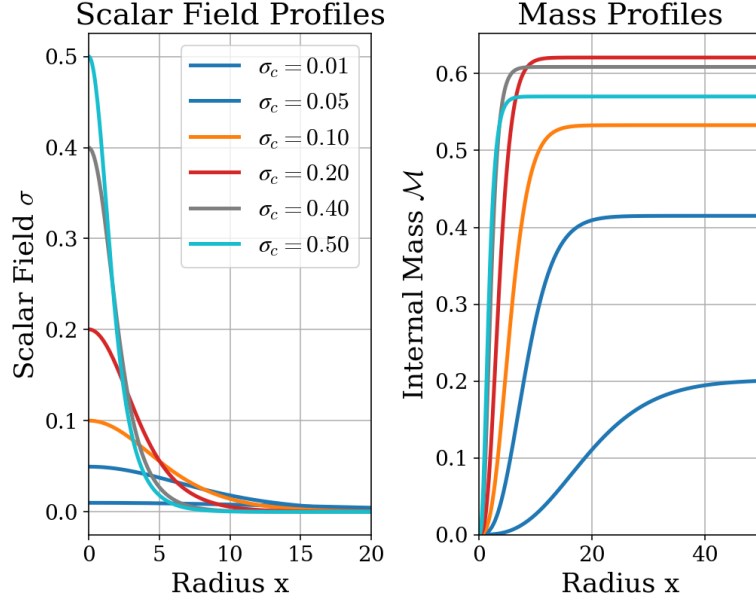


Figure 4.2: Boson star profiles (dimensionless scalar fields & masses) for various central field values σ_c at $\Lambda = 0$

The scalar field profiles in Figure 4.2 demonstrate that a higher central field value σ_c elevates the starting point of the profile and drives a significantly steeper decay, falling much more rapidly toward zero. Consequently, an increasing σ_c results in a more compact configuration with matter tightly concentrated near the core.

Similarly, as σ_c increases, the dimensionless internal mass rises more sharply with the radius, reaching its asymptotic limit of the dimensionless total mass $\mathcal{M}$ closer to the core, further confirming that the star becomes more compact. However, the total mass exhibits a non-monotonic behavior. Initially, increasing σ_c drives the total mass $\mathcal{M}$ upward, stabilizing at a higher value. Beyond a certain threshold, further increases in σ_c cause $\mathcal{M}$ to decrease, with the mass stabilizing at lower values.

Table 4.3: Summary of macroscopic boson star properties varying σ_c ($\Lambda = 0$)

Central Field σ_c	Nodes	Eigenfrequency Ω	Mass M [M_{Pl}^2/m]	Particles N [M_{Pl}^2/m^2]
0.01	0	0.993	0.202	0.202
0.05	0	0.967	0.415	0.419
0.10	0	0.937	0.533	0.543
0.20	0	0.884	0.621	0.639
0.40	0	0.809	0.609	0.624
0.50	0	0.785	0.570	0.575

The numerical data in Table 4.3 confirm this behavior. Up to $\sigma_c = 0.2$, each increase in σ_c yields a larger total mass $\mathcal{M}$. Beyond this value, the mass declines. The total number of particles $\mathcal{N}$ mirrors this behavior, peaking at $\sigma_c = 0.2$ before diminishing at higher σ_c .

This interesting behavior is explored thoroughly in Section 4.3, where the relationship between the total mass $\mathcal{M}$ and the central field σ_c is mapped continuously.

4.2.2 Parameter: Self-Interaction Strength Λ

The impact of the self-interaction strength, Λ , on the boson star configuration is evaluated by varying its value while holding the central field parameter constant ($\sigma_c = \text{const.}$). The following results pertain to a boson star with $\sigma_c = 0.1$. Figure 4.3 depicts the scalar field and mass profiles across a range of self-interaction values. These diagrams are complemented by Table 4.4, which records the corresponding macroscopic properties for each configuration: the eigenfrequency Ω , the total mass $\mathcal{M}$ and the total number of constituent particles $\mathcal{N}$.

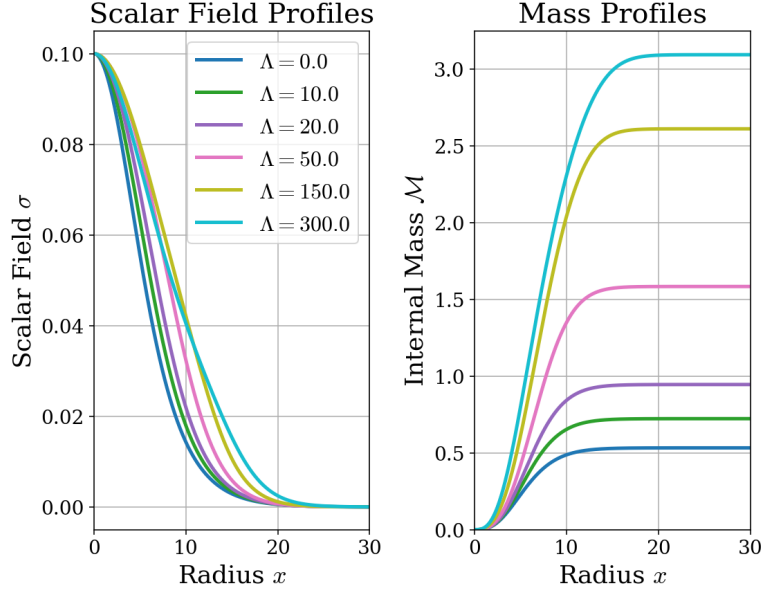


Figure 4.3: Boson star profiles (dimensionless scalar fields & masses) for various values of self-interaction Λ at $\sigma_c = 0.1$

The scalar field profiles in Figure 4.3 show that as the self-interaction strength Λ increases, the field approaches zero at progressively larger radii x . Nonetheless, this shift can no longer be associated with a simple uniform expansion. Instead, a distinct crossing behavior occurs where configurations with larger Λ initially exhibit a steeper decay near the core, causing them to drop below the curves of smaller Λ . Farther out, the slope of these high- Λ profiles flattens, causing them to cross back above the lower- Λ configurations. Consequently, profiles with smaller Λ reach zero at shorter radial distances, despite starting with the shallower core slopes. This behavior is particularly prominent with the profiles for $\Lambda = 150$ and $\Lambda = 300$, where the latter remains strictly lower until $x \approx 15$ before crossing over.

Plotting the spatial derivatives of the scalar field—or slope profiles—for these configurations in Figure 4.4 enables a more rigorous analysis. The diagram reveals that the behavior is even more complicated than initially observed, highlighting highly complex underlying dynamics. Apart from the tails, where smaller values of Λ guarantee a more rapid decay, there is no distinct interval where the magnitude of Λ monotonically dictates a faster or slower decay rate. Overall, the definitive conclusion is that an increasing self-interaction strength expands the effective extent of the scalar field.

The mass profiles in Figure 4.3 indicate that an increasing self-interaction strength yields a

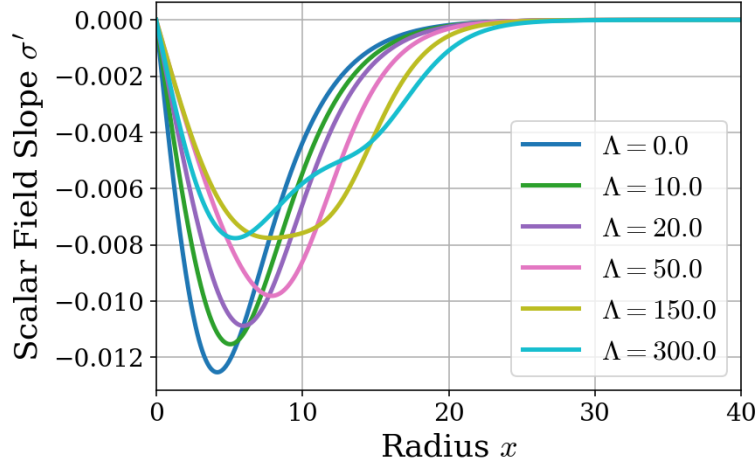


Figure 4.4: Scalar field slope profiles for various values of Λ at $\sigma_c = 0.1$

substantially larger total mass $\mathcal{M}$. However, mirroring the behavior of the scalar field, stabilization occurs farther from the core at larger values of x . Collectively, strong self-interactions support more massive stars by physically expanding them.

Table 4.4: Summary of profile results varying Λ ($\sigma_c = 0.1$)

Macroscopic Properties				
Λ	Nodes	Eigenfrequency Ω	Mass M [M_{Pl}^2/m]	Particles N [M_{Pl}^2/m^2]
0.0	0	0.937	0.533	0.543
10.0	0	0.922	0.723	0.743
20.0	0	0.905	0.945	0.980
50.0	0	0.860	1.584	1.673
150.0	0	0.803	2.611	2.763
300.0	0	0.797	3.094	3.172

The numerical data in Table 4.4 agree with the previous qualitative observations. As Λ scales up, the total mass $\mathcal{M}$ increases monotonically. The total number of particles $\mathcal{N}$ mirrors this behavior, confirming that strong self-interactions accommodate a significantly greater number of constituting particles.

4.3 Total Mass-Central Field Relationship

The investigation into how the central scalar field influences the boson star profile demonstrated that the total mass does not exhibit a monotonic behavior relative to the central field. To explore this behavior more rigorously, the dimensionless total mass $\mathcal{M}$ is plotted continuously as a function of the central field σ_c while holding the self-interaction strength parameter constant ($\Lambda = \text{const.}$).

Figure 4.5 illustrates this dimensionless total mass-central field relationship for a "mini" boson star ($\Lambda = 0$), alongside the corresponding eigenfrequency curve. The "mini" boson star

configuration was specifically chosen to allow direct cross-referencing with the heuristic results derived in Section 1.1.

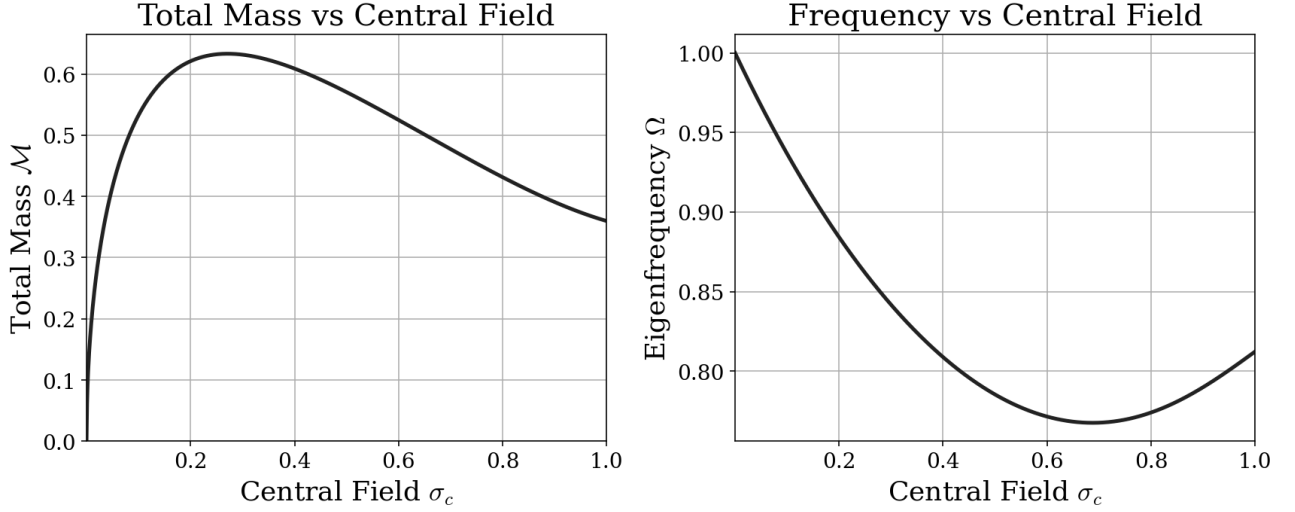


Figure 4.5: Dimensionless total mass & eigenfrequency relationship with central field at $\Lambda = 0$

The non-monotonic behavior of the total mass $\mathcal{M}$ relative to the central field σ_c is immediately evident in Figure 4.5. The mass $\mathcal{M}$ increases monotonically up to $\sigma_c = 0.271$, where it reaches a global maximum of $\mathcal{M}_{\max} = 0.633$ ($M_{\max} = 0.633 [M_{\text{Pl}}^2/m]$)—a value that is in complete agreement with established literature [3, 24, 2, 11, 25, 26]. Beyond this critical point, $\mathcal{M}$ decreases monotonically.

A maximal total mass of $M_{\max} = 0.633 [M_{\text{Pl}}^2/m]$ implies that all valid stellar masses for this configuration are of magnitude $10^{-1} [M_{\text{Pl}}^2/m]$. The heuristic arguments presented in Section 1.1 successfully predicted this exact order of magnitude, yielding a value only roughly 2.5 times smaller than the precise numerical result.

An important observation from the eigenfrequency curve is that all values strictly remain below 1, satisfying the theoretical constraint $\Omega \leq 1$ established in Section 3.1. Furthermore, $\Omega \rightarrow 1$ for very small central fields, which correctly aligns with the expected Newtonian limit [16, 2].

In the previous section, it was noted that the total number of particles $\mathcal{N}$ mirrors the behavior of the total mass $\mathcal{M}$. To examine this behavior more closely, Figure 4.6 plots both the total mass and the total number of particles together.

Figure 4.6 confirms the initial observation: the particle curve clearly shares the same monotonicity as the mass curve. Both values increase until $\sigma_c = 0.271$ and decrease thereafter. Furthermore, $\mathcal{N}$ and $\mathcal{M}$ are of the same order of magnitude. The two curves intersect, becoming equal ($\mathcal{M} = \mathcal{N}$) at $\sigma_c = 0.540$ —a crossing point that perfectly matches the bibliographical data [3, 11, 24]. Prior to this crossing point ($\sigma_c < 0.540$), the particle curve sits above the mass curve ($\mathcal{N} > \mathcal{M}$). After the crossing point ($\sigma_c > 0.540$) the particle curve drops below the mass curve ($\mathcal{N} < \mathcal{M}$). This shift is of immense physical importance, since it defines the sign of the binding energy, which in turn determines the energetic stability of the star.

The analysis concludes by investigating how the total mass-central field relationship is af-

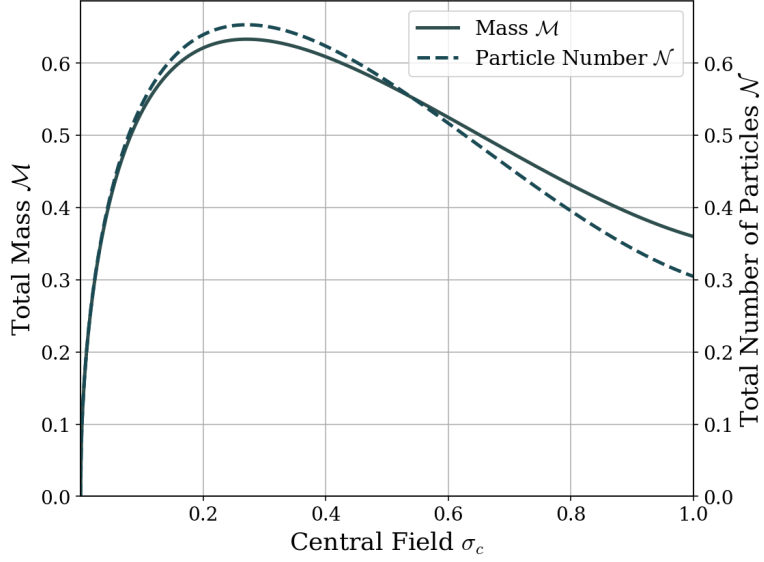


Figure 4.6: Total mass & total number of particles relationship with central field at $\Lambda = 0$

affected by the self-interaction strength parameter Λ . Figure 4.7 plots the mass and eigenfrequency curves for $\Lambda = 0, 1, 10, 30, 100, 200$ and 300 . These specific values were selected to match those used in Ref. [1], facilitating a direct comparison that confirms these results are identical to established literature.

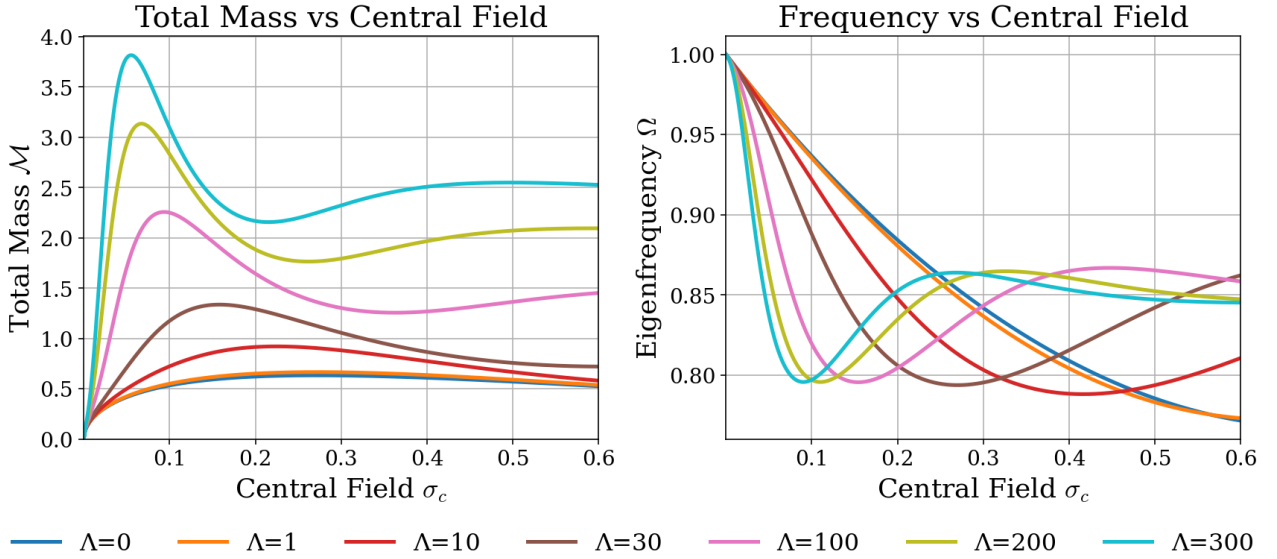


Figure 4.7: Total mass & eigenfrequency relationship with central field for various Λ

In Figure 4.7, it is apparent that the curves never intersect one another. This comes as no surprise; it was noticed in the previous section that an increase in Λ yields a larger total mass, and this holds true for every value of σ_c . In addition, as Λ increases, the peak of the curve becomes far more prominent and shifts towards the origin. In other words, a stronger self-interaction enables the star to support a greater maximum mass $\mathcal{M}_{\text{max}}$, but this maximum is reached at smaller σ_c values. This shifting trend is clearly shown in Figure 4.8, where the arrows indicate the direction of increasing Λ .

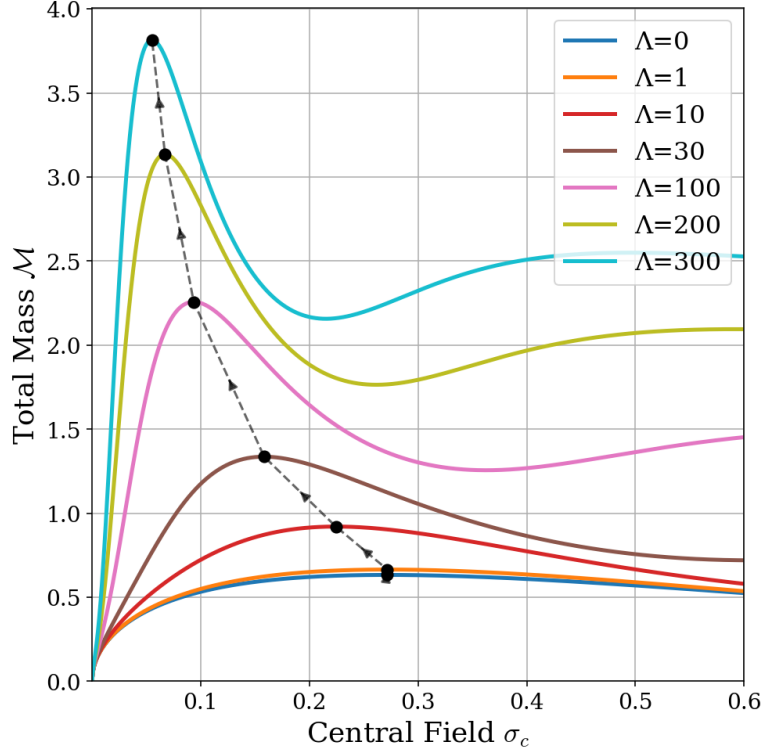


Figure 4.8: Maximum total mass position shift by increasing Λ

Finally, the curves corresponding to high- Λ values exhibit secondary extrema beyond their initial peak within this σ_c range. This does not imply that the low- Λ configurations lack these additional extrema. Their extrema are simply located at much larger σ_c values that fall outside the bounds of this plot. Since increasing the Λ shifts the entire curve toward the origin, the subsequent extrema for high- Λ are "pulled" within the plotted σ_c range.

Table 4.5 documents the exact coordinates of the first maximum of these curves: the maximum total mass $\mathcal{M}_{\max}$ and its corresponding central field σ_c . To provide a more intuitive understanding of the physical scale of these objects, these maximum masses are also calculated for a constituent boson of mass $m = 1$ [GeV].

Table 4.5: Maximum mass configurations and their corresponding physical masses for a constituent boson mass of $m = 1$ GeV.

Λ	$\sigma_c(M_{\max})$	$M_{\max} [M_{\text{Pl}}^2/m]$	$M_{\max, \text{phys}} [10^{-19} M_{\odot}]$	$M_{\max, \text{phys}} [10^{11} \text{ kg}]$
0.0	0.271	0.633	0.846	1.683
1.0	0.272	0.665	0.889	1.768
10.0	0.225	0.921	1.231	2.449
30.0	0.158	1.336	1.786	3.553
100.0	0.094	2.257	3.018	6.002
200.0	0.067	3.135	4.192	8.337
300.0	0.055	3.816	5.102	10.148

4.4 Fitting the Maximum Mass— Λ Relation

The prior analysis of the mass-central field curve demonstrated that an increase in the self-interaction strength Λ leads to a larger maximum mass $\mathcal{M}_{\text{max}}$. This trend has a precise quantitative form. Analytical calculations have established that the maximum boson star mass scales as [1, 2]

$$M_{\text{max}} \approx 0.22\Lambda^{1/2} \frac{M_{\text{Pl}}^2}{m} \quad (4.1)$$

which holds asymptotically in the Thomas-Fermi limit where $\Lambda \gg 1$ [4].

Interestingly, when the potential coupling constant λ is reinstated by substituting $\Lambda = \lambda M_{\text{Pl}}^2/(4\pi m^2)$, Equation (4.1) takes the form [2, 4]

$$M_{\text{max}} \approx 0.062\lambda^{1/2} \frac{M_{\text{Pl}}^3}{m^2} \quad (4.2)$$

which is comparable to the Chandrasekhar mass for fermions (see Table 1.1) with an effective fermion mass $m_{\text{fermion}} \sim m/\lambda^{1/4}$ [2]. Consequently, in the Thomas-Fermi limit, a fourth-order self-interaction results in a maximum mass on the order of $\sqrt{\lambda} M_{\text{Pl}}^3/m^2$ [4, 6, 27].

It is worth noting that the analytical derivation for large Λ ($\Lambda \gg 1$) yields a well-defined effective equation of state $p = p(\rho)$ [1, 4]. Therefore, in the Thomas-Fermi limit, a boson star becomes physically equivalent to an isotropic fluid star [1, 4], and one may use the perfect fluid formalism to investigate the equilibrium configurations and assess their stability [28].

Rather than relying exclusively on analytical approximations, we determine this relationship by fitting the $(\Lambda, \mathcal{M}_{\text{max}})$ data points extracted from the numerical solutions. Figure 4.9 plots those data points. The theoretical expectation that $\mathcal{M}_{\text{max}} \sim \sqrt{\Lambda}$, combined with the parabolic appearance of the data points, indicate a fit of the form $\mathcal{M}_{\text{max}} = \sqrt{a \cdot \Lambda + b}$, or equivalently, $\mathcal{M}_{\text{max}}^2 = a \cdot \Lambda + b$, where a and b are the coefficients to be determined.

To calculate these coefficients, a linear regression is performed on the transformed data points $(x = \Lambda, y = \mathcal{M}_{\text{max}}^2)$, taking advantage of the linear relationship $y = a \cdot x + b$. The least squares method yields $a = +0.047$ and $b = +0.375$, giving the fitted equation

$$\mathcal{M}_{\text{max}} = \sqrt{0.047\Lambda + 0.375} \quad (4.3)$$

The fitted model is remarkably accurate. There is outstanding agreement between the numerical data points and the fitted curve: qualitatively from the perfect alignment of the data points with the fitted line, and quantitatively from the R^2 coefficient that differs from the ideal 1 at the sixth decimal point ($R^2 = 0.999994$). At the same time, for high- Λ configurations, the fitted line tracks the theoretical line almost precisely, with a relative error of less than 1%.

This asymptotic behavior is easily verified algebraically. In the high- Λ regime ($\Lambda \gg 1$), the Λ term dominates ($0.047\Lambda \gg 0.375$) allowing the offset to be neglected. Taking this into account, the equation simplifies to $\mathcal{M}_{\text{max}} \approx \sqrt{0.047}\sqrt{\Lambda} \approx 0.217\sqrt{\Lambda}$, which perfectly recovers the analytical approximation $\mathcal{M}_{\text{max}} \approx 0.22\sqrt{\Lambda}$.

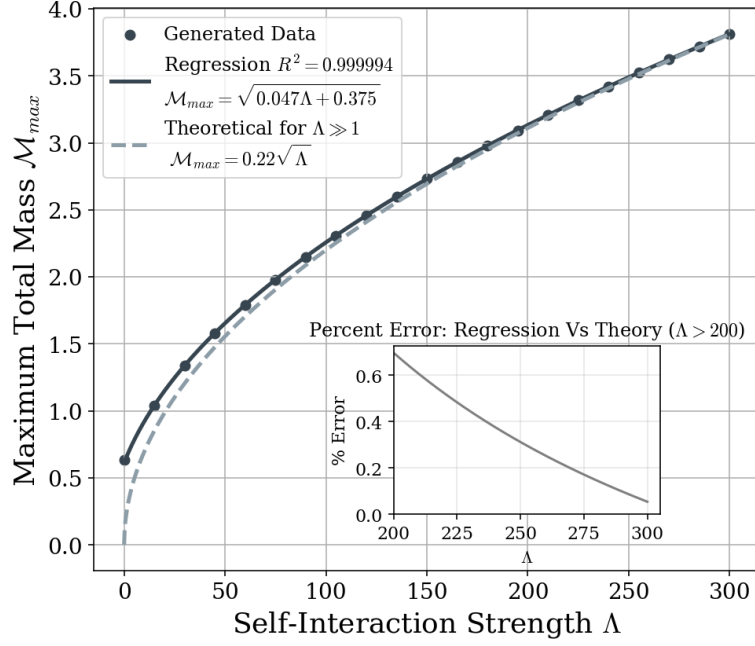


Figure 4.9: Maximum total mass relative to Λ : Regression & comparison with theoretical curve

4.5 solve_bvp Solver & Shooting Method Comparison

This chapter concludes with a preliminary comparison of the two numerical methods used to generate the data: Python’s built-in `solve_bvp` solver and the shooting method. This evaluation is based on two criteria: computational accuracy and execution efficiency.

Beginning with the accuracy evaluation, both methods generated identical plots and numerical data up to the selected three-decimal precision. To demonstrate this, the results of both solvers are plotted together, followed by an analysis of the differences between their calculated values. For brevity, this direct comparison is shown only for the total mass-central field relationship and the eigenfrequency-central field relationship.

Figure 4.10 displays on the top row the mass and eigenfrequency curves relative to the central field for both methods. It is apparent that both curves coincide perfectly, with no visible differences. The second row of the figure plots the exact numerical difference between the two datasets. The resulting residuals are on the order of 10^{-5} for the mass and 10^{-7} for the eigenfrequency. Both are negligible compared to the 10^{-1} baseline magnitude of these physical quantities.

Regarding execution efficiency, each method presents distinct advantages and disadvantages. The `solve_bvp` collocation algorithm is exceptionally fast, but highly sensitive to the solver parameters. It requires extensive fine-tuning of the integration intervals and number of mesh nodes before it successfully converges. However, once it is correctly calibrated, its computational speed is impressively high.

Conversely, the shooting method is incredibly versatile and robust, but computationally expensive. Once a boundary interval $[\Omega_{low}, \Omega_{high}]$ is established, which is typically straightforward, the solver reliably converges to a solution. However, because each iteration of the bisection

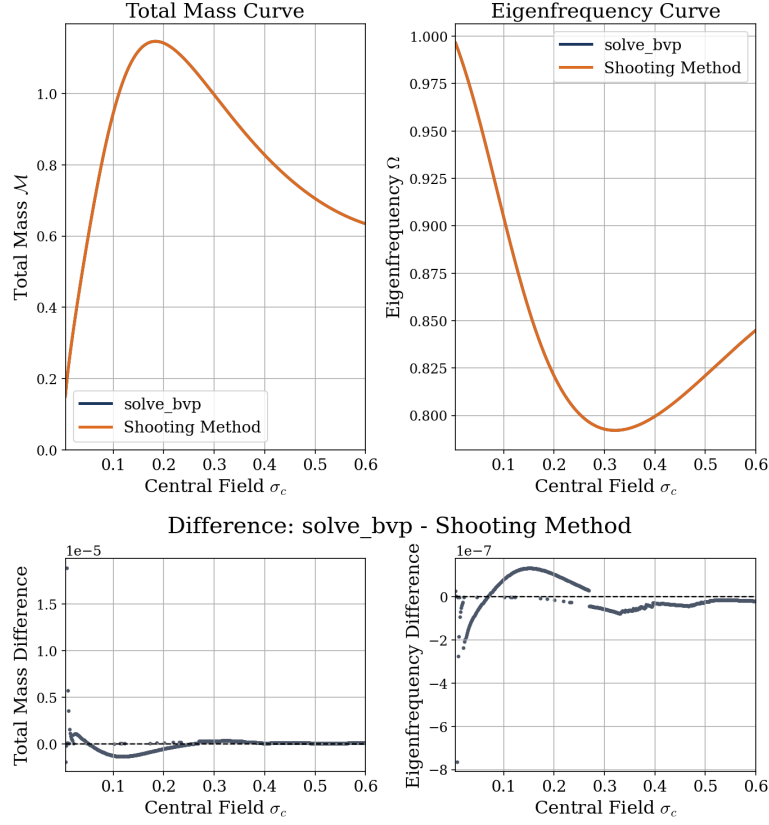


Figure 4.10: Comparison of the mass curve & eigenfrequency curve relative to the central field for the `solve_bvp` solver & the shooting method

algorithm requires solving a demanding initial value problem, achieving an accurate solution requires considerably more time.

To further quantify this substantial difference in performance between the two methods, Table 4.6 documents the execution times for three distinct parameter sweeps:

1. Calculating a single profile for the parameter set $\sigma_c = 0.1$ and $\Lambda = 0$.
2. Calculating profiles for 1001 values of σ_c equally distributed in the $[0, 1]$ interval for $\Lambda = 0$, to generate the mass curve.
3. Calculating profiles for 201 values of σ_c equally distributed in the $[0, 0.4]$ interval across four different self-interactions: $\Lambda \in \{0, 10, 100, 300\}$.

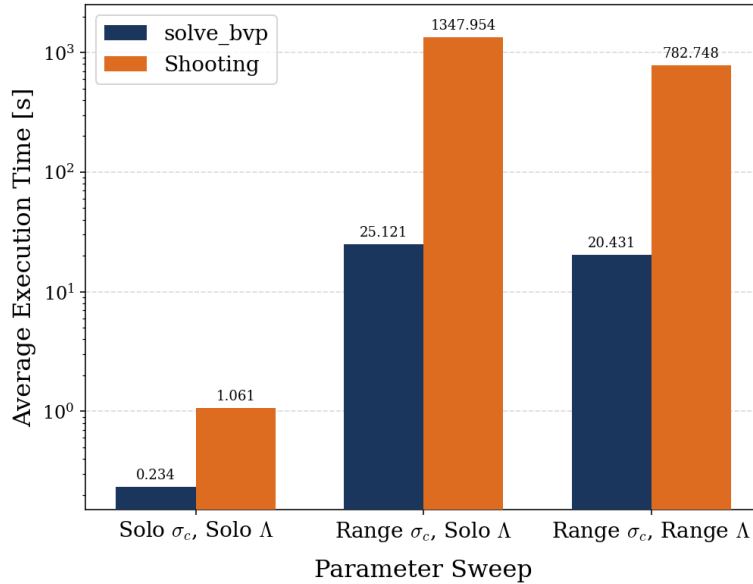
In each case, 61 iterations of the bisection algorithm are performed per profile. Consequently, the shooting method executes 61 evaluations in the first process, $61 \cdot 1001 = 61061$ evaluations in the second sweep and $61 \cdot 201 \cdot 4 = 49044$ evaluations in the third sweep. Because the third sweep requires fewer total evaluations than the second, its comparatively shorter execution time should come as no surprise.

Figure 4.11 visualizes the mean execution times of each method across the three sweeps. The immediate observation is that `solve_bvp` is significantly faster than the shooting method in every case. More importantly, however, this runtime difference does not scale linearly. For a single configuration, the execution time of the shooting method is roughly 4.5 times that of

Table 4.6: Comparison of execution times across different sweep cases

Execution Time [s]					
$\sigma_c = 0.1, \Lambda = 0$		$\sigma_c \in [0, 1], \Lambda = 0$		$\sigma_c \in [0, 0.4], \Lambda \in \{0, 10, 100, 300\}$	
solve_bvp	Shooting	solve_bvp	Shooting	solve_bvp	Shooting
0.246	1.066	25.051	1325.177	19.981	781.349
0.229	1.089	24.515	1387.604	20.793	795.138
0.240	1.070	24.095	1366.766	20.016	788.496
0.220	1.026	25.621	1334.825	19.816	776.555
0.234	1.053	26.323	1325.401	21.551	772.201
Avg	0.234	1.061	25.121	20.431	782.748

`solve_bvp`; yet, for the massive second sweep, it is more than 50 times slower. A final observation is that the execution time depends heavily on the total number of evaluations/repetitions rather than the complexity of varying multiple Λ values, as evidenced by the shorter runtime of the third sweep compared to the second.

Figure 4.11: Comparison of execution time of `solve_bvp` & the shooting method

Overall, for single configurations, the shooting method is preferable because of its reliable convergence and minimal fine-tuning. However, in extensive parameter sweeps, the `solve_bvp` solver becomes necessary to ensure computational efficiency and drastically reduce execution time.

Chapter 5

Gravitational Stability of Boson Stars

Boson stars, similar to other self-gravitating stellar objects, exhibit both stable and unstable configurations [29]. Earlier chapters focused on general equilibrium solutions of the Einstein-Klein-Gordon (EKG) equations. This chapter transitions from discussing mere numerical solutions to isolating the stable configurations that represent physically viable boson stars.

Because equilibrium solutions of boson stars are defined by a single parameter—the central scalar field, σ_c —stability theorems dictate that their dynamical fate can be deduced from their dimensionless total mass and particle number functions, $\mathcal{M} = \mathcal{M}(\sigma_c)$ and $\mathcal{N} = \mathcal{N}(\sigma_c)$ [2]. Section 5.1 analyses these curves to identify which configurations are dynamically stable (stable under radial perturbations). This section also evaluates the energetic stability of boson stars (their stability against complete dispersion into free constituent particles). Together, these analyses allow us to draw conclusions about the possible evolutionary endpoints of these equilibrium configurations. Finally, Section 5.2 draws parallels with neutron stars, using the mass-radius (M-R) diagram as an alternative way to determine gravitational stability.

5.1 Dynamical & Energetic Stability of Boson Stars

The stability of boson stars is evaluated across two distinct aspects: dynamical stability and energetic stability.

5.1.1 Dynamical Stability

Dynamical stability assesses the resistance of a boson star’s equilibrium configuration to radial perturbations [2]. There are two primary approaches for conducting dynamical stability analysis: linearly, by keeping the number of particles fixed ($\delta\mathcal{N} = 0$) and restricting the perturbation to their redistribution [24, 2]; and non-linearly, by permitting the number of particles to change ($\delta\mathcal{N} \neq 0$) either through accretion or annihilation [25, 2].

For stable configurations, linear stability analysis ($\mathcal{N} = \text{const.}$) predicts that a perturbed star will oscillate and eventually return to its exact initial state [24]. Conversely, non-linear

analysis suggests it will emit scalar radiation and ultimately settle into a new configuration with a smaller total mass [25, 2]. On the other hand, unstable configurations will either collapse into black holes or disperse away, regardless of the approach [2, 24, 11, 29]. A unique result of non-linear analysis ($\delta\mathcal{N} \neq 0$) is the potential migration of an unstable configuration to a stable one, depending on the nature of the initial perturbation. The remainder of this section focuses primarily on linear stability analysis, with occasional remarks regarding the non-linear case where relevant.

In linear stability analysis, radial perturbations assume a harmonic time dependence with frequency χ . Combined with the constraint $\delta\mathcal{N} = 0$, this yields an eigenvalue equation for normal modes—specifically for χ^2 [24, 2]. A configuration is stable only when $\chi^2 > 0$ across all modes (the frequencies are real), and it is unstable if any mode produces $\chi^2 < 0$ (an imaginary frequency that dictates exponential growth) [24, 2]. Because the fundamental mode is the first to go unstable, the analysis is reduced to evaluating χ_0^2 . If $\chi_0^2 > 0$, all modes are stable [24]. If $\chi_0^2 < 0$, the fundamental mode grows exponentially with time, rendering the star unstable [2]. Consequently, the boundary between stability and instability corresponds to zero frequency perturbations, where $\chi_0^2 = 0$ [24].

Equilibrium configurations are parameterized solely by the central scalar field, σ_c . Thus, the dimensionless total mass $\mathcal{M}$ and total particle number $\mathcal{N}$ are functions of σ_c , $\mathcal{M} = \mathcal{M}(\sigma_c)$ and $\mathcal{N} = \mathcal{N}(\sigma_c)$ [2]. In a zero frequency perturbation, the vanishing eigenvalue causes the harmonic time dependence to vanish as well. Therefore, perturbing an equilibrium configuration at σ_c simply results in a neighboring equilibrium configuration at $\sigma_c + \delta\sigma_c$, for infinitesimal $\delta\sigma_c$ [24]. Since the total particle number must be conserved ($\delta\mathcal{N} = 0$), zero frequency perturbations exist only at critical points where $d\mathcal{N}/d\sigma_c = 0$. It is important to emphasize that a zero frequency perturbation does not necessarily correspond to the fundamental mode; it may represent a higher mode crossing zero.

In summary, transitions between stability and instability occur only at critical points where [2, 24]

$$\frac{d\mathcal{N}}{d\sigma_c} = 0 = \frac{d\mathcal{M}}{d\sigma_c} \quad (5.1)$$

However, a critical point is a necessary but not sufficient condition for a change in the star's stability; instead it could indicate a higher mode going unstable [24].

Ref. [24] proves that the first critical point—associated with the maximum total mass $\mathcal{M}_{\max}$ and maximum particle number $\mathcal{N}_{\max}$ —corresponds to the fundamental mode χ_0^2 going from positive to negative, marking the definitive transition of the boson star from stable to unstable. Subsequent extrema correspond to higher modes sequentially turning negative; however, this does not alter the stability of the star, which remains unstable [30].

This analysis establishes an important conclusion: a boson star is stable in the region preceding the global maximum of $\mathcal{M}_{\max}$ and $\mathcal{N}_{\max}$ ($\sigma_c \leq \sigma_{c(\max)}$), known as the S-branch, and it is unstable in the region following the global maximum ($\sigma_c \geq \sigma_{c(\max)}$), known as the U-branch [29, 25]. Therefore, the maximum total mass serves as the dynamical stability boundary for the boson star [3].

Figure 5.1 clearly illustrates these branches for a "mini" boson star ($\Lambda = 0$). The maximum total mass $\mathcal{M}_{\max}$ and maximum particle number $\mathcal{N}_{\max}$ are located at $\sigma_{c(\max)} = 0.271$. The region to the left of the maximum ($\sigma_c < 0.271$) is the S-branch where the boson star is stable, while the region to the right ($\sigma_c > 0.271$) is the U-branch where the boson star is unstable.

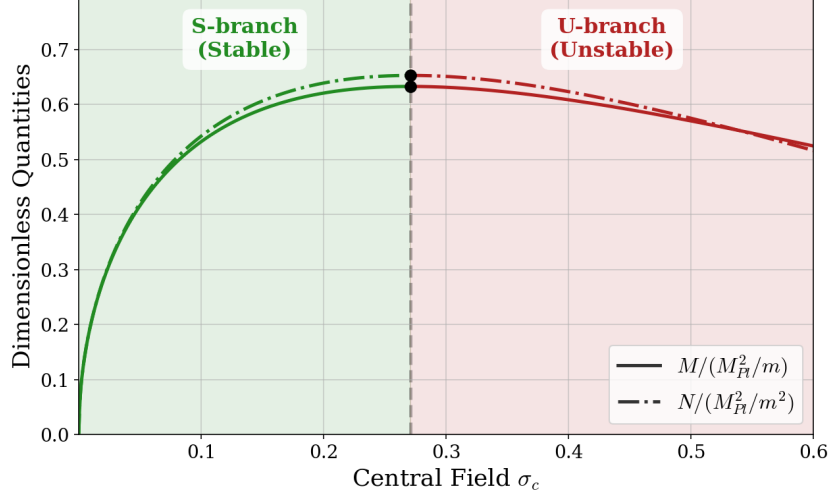


Figure 5.1: Dynamical stability & instability regions at $\Lambda = 0$

Finally, the results of the non-linear stability analysis agree with those of the linear analysis, confirming that the transition from stable to unstable occurs exactly at the critical point of $\mathcal{M}_{\max}$ and $\mathcal{N}_{\max}$.

5.1.2 Energetic Stability

Energetic stability determines whether an equilibrium configuration is gravitationally bound, allowing it to exist as a stellar object, or it will ultimately disperse into free particles at infinity [11, 2]. This characteristic is evaluated using the binding energy E_b defined as [2, 3, 24, 11, 29, 25]

$$E_b = M - N \cdot m \quad (5.2)$$

where M represents the total mass of the star (in M_{Pl}^2/m units), N is the total number of particles (in M_{Pl}^2/m^2 units) and m is the mass of the constituent boson.

Substituting the rescaled dimensionless expressions for mass, $\mathcal{M} = M/(M_{\text{Pl}}^2/m)$, and particle number, $\mathcal{N} = N/(M_{\text{Pl}}^2/m^2)$, into Equation (5.2) yields

$$E_b = \mathcal{M} \frac{M_{\text{Pl}}^2}{m} - \mathcal{N} \frac{M_{\text{Pl}}^2}{m^2} m = (\mathcal{M} - \mathcal{N}) \frac{M_{\text{Pl}}^2}{m} \quad (5.3)$$

By defining the rescaled dimensionless binding energy as $\mathcal{E}_b = E_b/(M_{\text{Pl}}^2/m)$, we arrive at the relation

$$\mathcal{E}_b = \mathcal{M} - \mathcal{N} \quad (5.4)$$

For a configuration to be bound, the total mass of the system must be strictly less than the aggregate mass of all the constituent bosons ($M < N \cdot m$, or equivalently $\mathcal{M} < \mathcal{N}$) yielding a

negative binding energy ($E_b < 0$, or $\mathcal{E}_b < 0$). Conversely, an unbound configuration possesses an excess of energy [2], where the mass of the system exceeds the aggregate particle mass ($M > N \cdot m$, or $\mathcal{M} > \mathcal{N}$). As a result, the binding energy is positive ($E_b > 0$, or $\mathcal{E}_b > 0$), indicating that no external work is required to disassemble the star [29]. The excess energy translates into kinetic energy, driving the dispersion of free bosons at infinity. Therefore, the boundary defining energetic stability—where the boson star transitions from a bound to an unbound state—occurs where the binding energy is zero ($\mathcal{E}_b = 0$, or equivalently $\mathcal{M} = \mathcal{N}$).

Figure 5.2 maps these regions for a "mini" boson star ($\Lambda = 0$). The transition point of zero binding energy ($\mathcal{E}_b = 0$), corresponding to the intersection of the mass and particle number curves, is located at $\sigma_{c(\mathcal{E}_b=0)} = 0.540$. The region to the left of this threshold ($\sigma_c \leq 0.540$) represents bound configurations, while the region to the right ($\sigma_c \geq 0.540$) denotes unbound configurations.

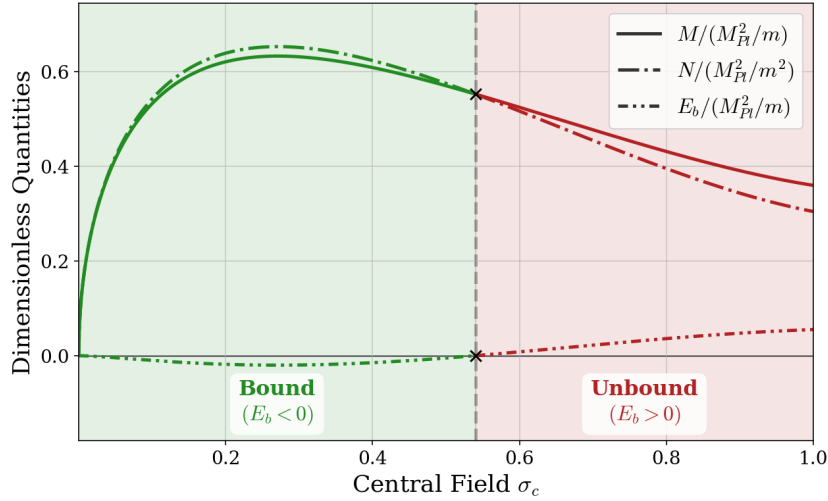


Figure 5.2: Energetic stability (bound) & instability (unbound) regions at $\Lambda = 0$

5.1.3 The Three Evolutionary Fates of Boson Stars

The preceding sections establish two points of interest regarding the gravitational stability of a boson star [29]:

- The dynamical stability boundary is defined by the critical point of maximum mass, $\mathcal{M}_{\max}$ (where $\sigma_c = \sigma_{c(\max)}$). Equilibrium configurations in the region to the left of this threshold are stable, while those to its right are unstable.
- The energetic stability boundary corresponds to the point where the binding energy is zero, $\mathcal{E}_b = \mathcal{M} - \mathcal{N} = 0$ (where $\sigma_c = \sigma_{c(\mathcal{E}_b=0)}$). Equilibrium configurations in the region to the left of this point are gravitationally bound ($\mathcal{E}_b < 0$), while those to its right are unbound ($\mathcal{E}_b > 0$).

Since these two boundaries do not coincide (strictly $\sigma_{c(\max)} < \sigma_{c(\mathcal{E}_b=0)}$), they partition the parameter space into three distinct regions. For visual aid, Figure 5.3 maps these two boundaries for the "mini" boson star case ($\Lambda = 0$): $\sigma_{c(\max)} = 0.271$ for the critical point of $\mathcal{M}_{\max}$,

and $\sigma_{c(\mathcal{E}_b=0)} = 0.540$ for the zero binding energy threshold. The type of stability associated with each region is documented according to the aforementioned criteria.

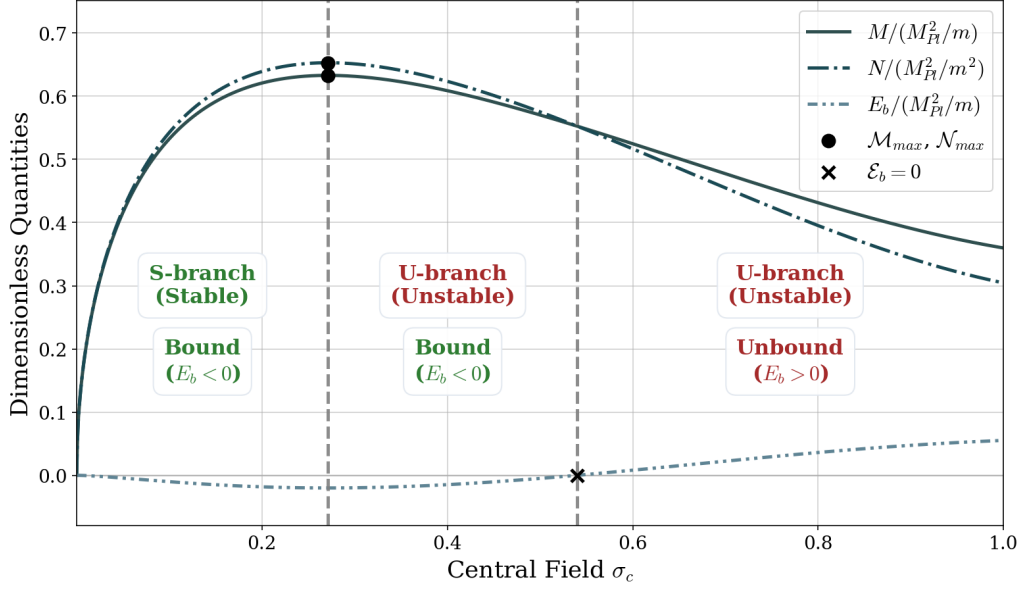


Figure 5.3: Points of interest for stability at $\Lambda = 0$

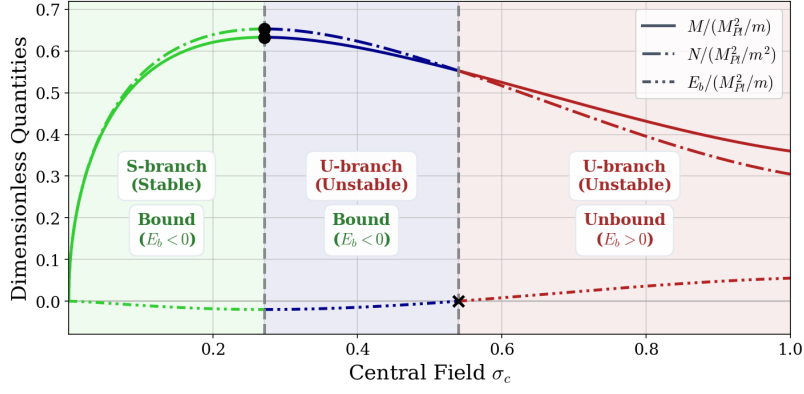
These boundaries define three specific evolutionary regions:

1. $\sigma_c < \sigma_{c(\max)}$: Equilibrium configurations here are both stable and bound [29, 2]. This is the only region where the configurations represent physically viable boson stars.
2. $\sigma_{c(\max)} < \sigma_c < \sigma_{c(\mathcal{E}_b=0)}$: Equilibrium configurations here are dynamically unstable, but remain energetically bound. In this intermediate region, the configurations will ultimately collapse into a black hole [29, 2].
3. $\sigma_c > \sigma_{c(\mathcal{E}_b=0)}$: Equilibrium configurations here are both unstable and unbound. In this region, the constituent bosons disperse to infinity, where the excess energy is converted to kinetic energy of the free particles [29, 2].

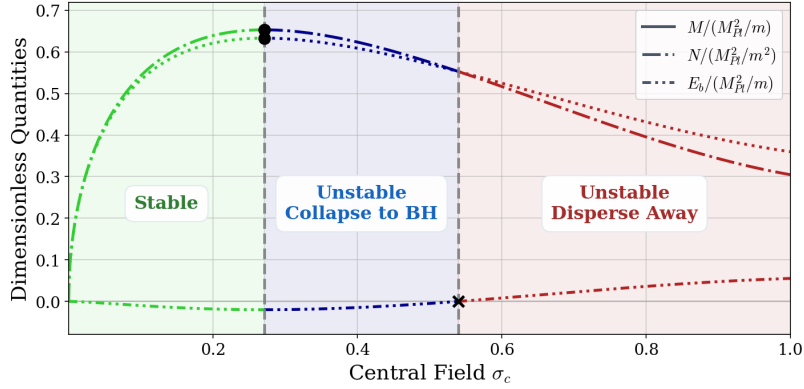
Figure 5.4a presents a color-coded mapping of these regions. While it does not introduce new data, it serves as a visual bridge to the evolutionary endpoints depicted in Figure 5.4b.

Figure 5.4b explicitly links these regions with the three terminal fates of a "mini" boson star equilibrium configuration ($\Lambda = 0$). For $\sigma_c < 0.271$, the configurations persist as stable boson stars. For $0.271 < \sigma_c < 0.540$, configurations will undergo gravitational collapse into a black hole. Finally, for $\sigma_c > 0.540$, the configuration will completely disperse away. These conclusions are in complete agreement with established literature [11, 29, 2].

Notably, non-linear stability analysis introduces an additional evolutionary scenario: the migration of an equilibrium configuration between the U-branch and S-branch. If a radial perturbation is induced by particle annihilation, an unstable U-branch configuration may cross the dynamical stability boundary, enter the stable region and—facilitated by a damping mechanism—settle into a stable S-branch state. Conversely, if the radial perturbation is



(a) Color-coded stability regions



(b) Evolutionary fates

Figure 5.4: Definition of stability regions & their corresponding evolutionary fates at $\Lambda = 0$

caused by particle accretion, such a migration is impossible. On the contrary, this type of radial perturbation may drive a stable S-branch configuration to migrate into the unstable U-branch.

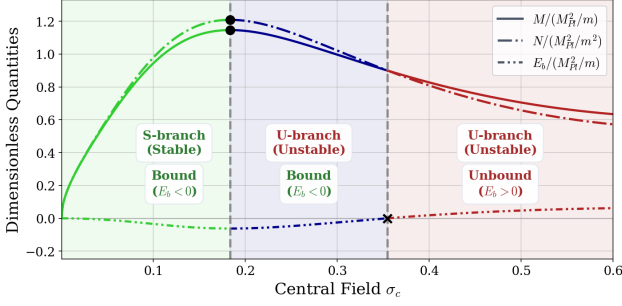
5.1.4 Generalization

The conclusions regarding how the three stability regions are defined and their corresponding evolutionary fates are fully generalizable across any self-interaction strength, Λ . The initial analysis focused on the non-interacting "mini" boson star ($\Lambda = 0$) because it is extensively documented in the literature, allowing for straightforward verification of the methodology. For completeness, the stability criteria are applied to self-interacting configurations with $\Lambda = 20$ and $\Lambda = 100$.

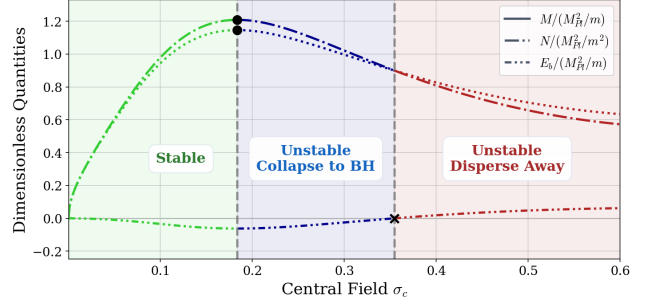
For a self-interaction strength of $\Lambda = 20$, the critical point of $\mathcal{M}_{\max}$ is located at $\sigma_{c(\max)} = 0.184$. This dictates that for $\sigma_c < 0.184$ the configurations are stable, whereas for $\sigma_c > 0.184$ they are unstable.

The zero binding energy threshold ($\mathcal{E}_b = 0$) is located at $\sigma_{c(\mathcal{E}_b=0)} = 0.354$. Consequently, for $\sigma_c < 0.354$ the configurations are gravitationally bound, while for $\sigma_c > 0.354$ they are unbound.

Figure 5.5a clearly marks these regions. Following the established criteria, configurations for $\sigma_c < 0.184$ are stable boson stars. For $0.184 < \sigma_c < 0.354$, the configurations undergo gravitational collapse into a black hole. Finally, for $\sigma_c > 0.354$, the configurations disperse to free bosons at infinity. These terminal fates are illustrated in Figure 5.5b.



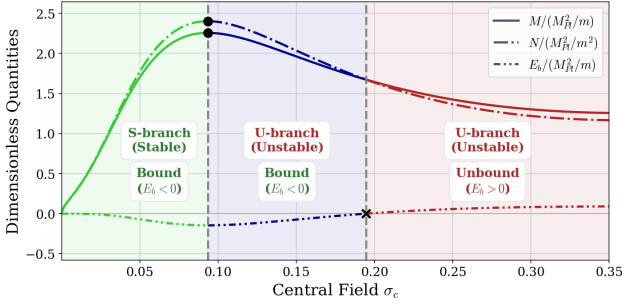
(a) Color-coded stability regions



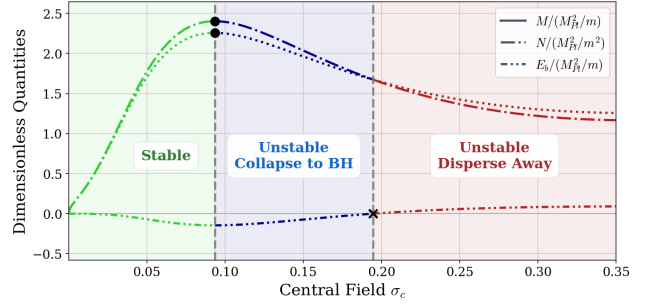
(b) Evolutionary fates

Figure 5.5: Definition of stability regions & their corresponding evolutionary fates at $\Lambda = 20$

For a self-interaction of $\Lambda = 100$, the dynamical stability boundary ($\mathcal{M}_{\max}$) and the energetic stability boundary ($\mathcal{E}_b = 0$) occur at $\sigma_{c(\max)} = 0.094$ and $\sigma_{c(\mathcal{E}_b=0)} = 0.195$, respectively. Therefore, as mapped in Figure 5.6, configurations are stable for $\sigma_c < 0.094$, collapse into a black hole in the intermediate region $0.094 < \sigma_c < 0.195$ and disperse away when $\sigma_c > 0.195$.



(a) Color-coded stability regions



(b) Evolutionary fates

Figure 5.6: Definition of stability regions & their corresponding evolutionary fates at $\Lambda = 100$

5.2 Parallels with Neutron Stars: Mass-Radius Relationship

The macroscopic properties and stability criteria of boson stars present profound similarities to those of neutron stars [25, 26]. In both cases, the equilibrium configurations are parameterized by a central concentration variable: central density, ρ_c , for neutron stars and the central value of the field, σ_c , for boson stars. Consequently, the total mass can be expressed as a function of this central parameter alone: $\mathcal{M} = \mathcal{M}(\rho_c)$ and $\mathcal{M} = \mathcal{M}(\sigma_c)$, respectively. The resulting mass curve relative to the central field for boson stars exhibits clear qualitative similarities to that of a neutron star [25]. Furthermore, the stability theorem establishing that boson star configurations transition between stability and instability only at critical points where $d\mathcal{M}(\sigma_c)/d\sigma_c = 0$ directly mirrors the $d\mathcal{M}(\rho_c)/d\rho_c = 0$ criterion used in neutron stars [31].

These similarities motivate the construction of a mass-radius ($M - R$) diagram to analyze boson star stability, as is typically done in neutron stars [32, 33]. However, unlike neutron stars,

which possess a well-defined surface where fluid density drops discontinuously to zero, boson stars asymptotically extend to infinity and therefore lack a sharp boundary and a definitive physical radius [2, 26]. Nonetheless, due to the exponential decay of the scalar field and the asymptotic behavior of the internal mass, one conventionally defines an effective radius that contains a fixed percentage of the total stellar mass [2]. Adopting the convention of Ref. [25], the effective radius of the star, R , is henceforth defined as the radial distance encompassing 95% of the total mass.

Figure 5.7 illustrates the total mass $\mathcal{M}$ versus the effective radius R for a "mini" boson star ($\Lambda = 0$). The arrows indicate the direction of increasing central field, σ_c . The inset provides a magnified view of the characteristic spiral, "snail-like" behavior at small radii [25, 26, 4]. This $M - R$ diagram shares key similarities with those constructed for neutron stars [25, 26].

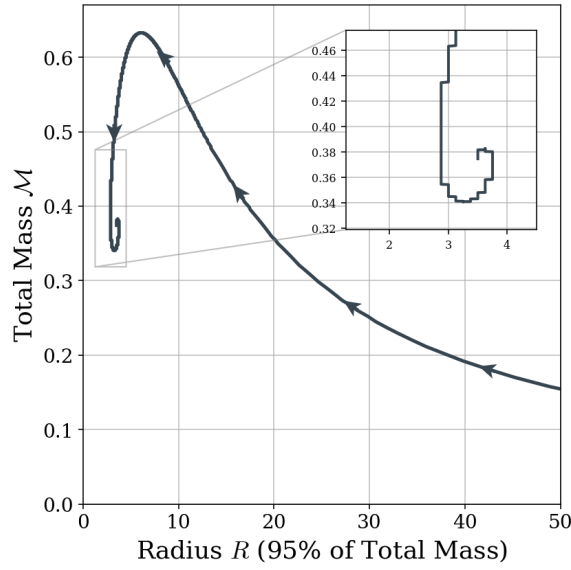


Figure 5.7: Total mass-effective radius (containing 95% of the total mass) diagram at $\Lambda = 0$

The stability criteria of normal radial modes for the $M - R$ diagram method are formulated as follows [32]:

1. **Changing stability:** Normal radial modes change stability exclusively at the critical points of the $\mathcal{M}(R)$ curve; at each critical point only one mode transitions between stability and instability.
2. **Direction of the curve bend and stability character:** One mode becomes unstable only if the $\mathcal{M}(R)$ curve bends counter-clockwise at the critical point; conversely, it transitions to a stable configuration only if the curve bends clockwise.

These criteria are applied to the $M - R$ diagram in Figure 5.7. Assuming the lowest density segment (corresponding to the largest radius) to be stable for all radial modes, and following the direction of the arrows, the fundamental mode becomes unstable at the first critical point ($\mathcal{M}_{\max}$), rendering the star unstable. Due to the inward spiral feature of the curve, it bends counter-clockwise at each subsequent critical point encountered in the direction of increasing

σ_c . This implies higher modes sequentially turn unstable at each turning point, ensuring the star remains unstable. Thus, the $\mathcal{M}(R)$ curve splits into a stable S-branch preceding $\mathcal{M}_{\max}$ and an unstable U-branch following it, reinforcing the previous findings.

Figure 5.8 isolates the S-branch and U-branch for a "mini" boson star ($\Lambda=0$). The stability boundary defined by the maximum total mass ($\mathcal{M}_{\max} = 0.633$) occurs at an effective radius of $R_{\max} = 6.011$. These results are consistent with both established literature [25] and our prior analysis.

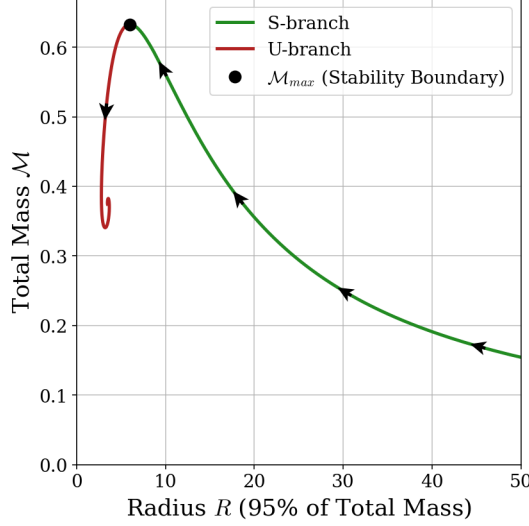


Figure 5.8: S-branch & U-branch on M-R diagram at $\Lambda = 0$

Expanding the analysis to the more general self-interacting case ($\Lambda \neq 0$), Figure 5.9 depicts the $M - R$ diagram, with its S- and U-branches clearly marked, for a self-interaction strength of $\Lambda = 100$. Qualitatively, the $M - R$ diagram preserves the characteristic spiral structure. Quantitatively, the stability boundary corresponding to $\mathcal{M}_{\max} = 2.257$ occurs at an effective radius $R = 13.253$. Consistent with the findings from Chapter 4, introducing a self-interaction drives both the maximum total mass and its corresponding radius upward.

To provide a comprehensive physical perspective, Table 5.1 details the stability boundary values for the previously discussed models ($\Lambda = 0$ and $\Lambda = 100$), alongside two additional self-interacting cases ($\Lambda = 20$ and $\Lambda = 300$). By restoring the fundamental constants, these results are evaluated for realistic constituent boson candidates. The mass is measured in units of M_{Pl}^2/m via $M = \mathcal{M} \cdot (M_{\text{Pl}}^2/m)$, while the radius is measured in units of $1/m$ via $R_{\text{phys}} = R/m$. Table 5.1 summarizes these maximum macroscopic values for constituent boson masses of 10^{-22} [eV] (ultra-light axions), 10^{-10} [eV] (QCD axions), and 1 [GeV] (WIMPs).

The values derived in Table 5.1 highlight the remarkably wide range of macroscopic masses and radii of boson stars, extending from sub-atomic dimensions to galactic scales, depending on the constituting particle mass. Furthermore, the data intuitively demonstrate how increasing the self-interaction increases both the mass and size of the boson star.

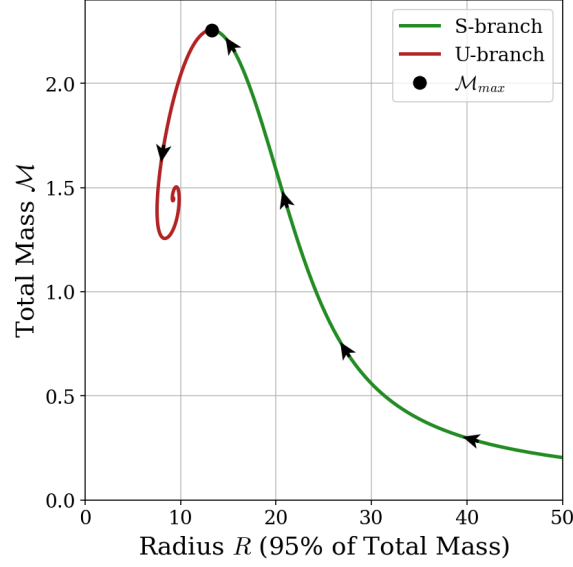


Figure 5.9: S-branch & U-branch on M-R diagram at $\Lambda = 100$

Λ	Stability Boundary		Ultra-light Axion 10^{-22} [eV]	QCD Axion 10^{-10} [eV]	Scalar WIMP 1 [GeV]
0	$M_{\max}$	$0.633 [M_{\text{Pl}}^2/m]$	$8.45 \times 10^{11} M_{\odot}$	$0.84 M_{\odot}$	1.68×10^{11} kg
	$R_{\max}$	$6.011 [1/m]$	0.38 pc	11.8 km	1.18 fm
20	$M_{\max}$	$1.147 [M_{\text{Pl}}^2/m]$	$1.53 \times 10^{12} M_{\odot}$	$1.52 M_{\odot}$	3.04×10^{11} kg
	$R_{\max}$	$7.739 [1/m]$	0.49 pc	15.2 km	1.52 fm
100	$M_{\max}$	$2.257 [M_{\text{Pl}}^2/m]$	$3.01 \times 10^{12} M_{\odot}$	$3.00 M_{\odot}$	5.99×10^{11} kg
	$R_{\max}$	$13.253 [1/m]$	0.84 pc	26.0 km	2.60 fm
300	$M_{\max}$	$3.816 [M_{\text{Pl}}^2/m]$	$5.09 \times 10^{12} M_{\odot}$	$5.06 M_{\odot}$	1.01×10^{12} kg
	$R_{\max}$	$21.399 [1/m]$	1.35 pc	42.0 km	4.20 fm

Table 5.1: Comparison between numerical results and physical dimensions of boson stars across different self-interaction strengths (Λ) for various dark matter candidates.

Chapter 6

Applications

This final chapter provides a brief overview of the potential applications of boson star configurations and the astrophysical roles they may fulfill. The most compelling scenarios involve their nature as compact objects, their ability to mimic the behavior of black holes, their candidacy for dark matter, and their function as novel gravitational wave sources.

The macroscopic properties summarized in Table 5.1 indicate that boson stars inherently possess the sizes and masses of typical astrophysical compact objects [5]. Therefore, they are often classified as exotic compact objects [5, 34]. It has been proposed that constituent bosons could condense into a compact object through a Jeans instability mechanism, [35], where the ground-state boson star represents the zero-temperature limit of the resulting self-gravitating Bose-Einstein condensate [29]. A defining characteristic of such an object is its lack of a Schwarzschild-type event horizon and singularities; it maintains a regular geometry extending to infinity and a smooth mass distribution [29, 4]. Another intriguing possibility, beyond isolated boson star configurations, is the coexistence of exotic bosonic matter within neutron stars [2].

As non-vacuum alternative solutions to black holes, boson stars could effectively mimic the behavior of black holes [34, 29]. In fact, given their potentially immense masses made possible by ultra-light bosons (as seen in Table 5.1), these configurations could accurately emulate even supermassive black holes. Crucially, however, boson stars lack an event horizon and a hard surface, which can lead to distinct observational behaviors [2].

Perhaps the most prominent astrophysical role a boson star could fulfill is accounting for the missing cosmological mass [1]. It has been proposed that dark matter could consist of scalar bosonic particles [25, 35], with the best-motivated candidate being the QCD axion [5, 36]. These axions would condense in a bound, macroscopic Bose-Einstein condensate to form an axion star [36].

Another scenario considers the presence of bosonic fields inside neutron stars, although this raises potential issues with observations of old neutron stars [2]. Furthermore, owing to their enormous radii in the Newtonian limit, boson stars could form galactic dark matter halos [29, 4], which would naturally explain the flattening of the galactic rotation curves.

Finally, gravitational wave detection offers a powerful tool in the search for boson stars. These

stellar objects could serve as sources of gravitational waves [11], with the most natural scenario being a boson star binary system [2, 5]. Evidence from future gravitational wave observations could confirm the existence of exotic compact objects, potentially providing definitive proof of dark matter particles. Moreover, while boson stars can imitate supermassive black holes, their lack of an event horizon and a hard surface is expected to produce distinguishable gravitational wave signatures.

Chapter 7

Conclusion

In this thesis, we have thoroughly investigated ground-state boson stars possessing a quartic self-interaction ($\frac{1}{4}\lambda|\phi|^4$), with particular emphasis on the limiting case of non-interacting "mini" boson stars. Using a variational approach within General Relativity, we derived the Einstein-Klein-Gordon equations that describe the system and subsequently solved them numerically using two distinct methodologies: Python's built-in `solve_bvp` solver and a one-parameter shooting method (utilizing Runge-Kutta-Fehlberg and bisection algorithms) applied to a rescaled system. Both numerical approaches yielded identical profiles and parameter curves, in excellent agreement up to at least five decimal places. The `solve_bvp` solver proved to be exceptionally fast and efficient, rendering it ideal for massive continuous parameter sweeps; however, it required meticulous fine-tuning to ensure convergence. Conversely, while the shooting method was computationally expensive and prohibitive for larger datasets, it provided a highly robust alternative that guaranteed convergence with minimal calibration.

Our physical analysis confirmed that the total mass of boson star configurations scales inversely with the mass of the constituent bosons ($M \sim M_{\text{Pl}}^2/m$). Numerically, we reproduced the Kaup limit for non-interacting "mini" boson stars, identifying a maximum mass of $0.633[M_{\text{Pl}}^2/m]$ at a central field value of $\sigma_c = 0.271$. Furthermore, the simulations revealed that increasing the central value of the scalar field, σ_c , concentrates matter more tightly near the core, producing a more compact star. It was also observed that increasing the self-interaction strength, Λ , expanded the effective extent of the scalar field, enabling the star to support significantly larger masses. A regression analysis of the maximum total mass of each configuration family as a function of its respective self-interaction strength Λ yielded the relationship $\mathcal{M}_{\text{max}} = \sqrt{0.047\Lambda + 0.375}$, which perfectly aligns with the Thomas-Fermi limit for large Λ ($\Lambda \gg 1$), where $\mathcal{M}_{\text{max}} \approx 0.22\Lambda^{1/2}$.

Finally, we determined that the dynamical and energetic stability of these equilibrium configurations depends entirely on the central scalar field parameter, σ_c , resulting in three distinct evolutionary fates. Configurations along the S-branch ($\sigma_c < \sigma_{c(\text{max})}$) are dynamically stable and gravitationally bound, representing the only physically viable boson stars. Configurations within the intermediate U-branch ($\sigma_{c(\text{max})} < \sigma_c < \sigma_{c(\mathcal{E}_b=0)}$) are dynamically unstable but gravitationally bound; therefore, they ultimately collapse into black holes. Configurations in the

extreme U-branch ($\sigma_c > \sigma_{c(\mathcal{E}_b=0)}$) are both dynamically unstable and unbound, meaning their excess energy will cause the configuration to completely disperse into free particles at infinity.

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